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is attributable to cardiovascular disease. Strict control of blood pressure and blood sugar and lipid levels is essential in this population.

Hypertension may be caused by (1) native kidney disease, (2) rejection activity in the transplant, (3) renal artery stenosis if an end-to-end anastomosis was constructed with an iliac artery branch, and (4) renal CNI toxicity, which may improve with reduction in dose. Whereas ACE inhibitors may be useful in the longer term, calcium channel blockers are more frequently used initially. Amelioration of hypertension to the range of 120–130/70–80 mmHg should be the goal in all patients.

Hypercalcemia after transplantation may indicate failure of hyperplastic parathyroid glands to regress. Aseptic necrosis of the head of the femur when it occurs is probably due to preexisting hyperparathyroidism, with aggravation by glucocorticoid treatment. With improved management of calcium and phosphorus metabolism during chronic dialysis, the incidence of parathyroid-related complications has fallen dramatically. Persistent hyperparathyroid activity may require subtotal parathyroidectomy.

Although most transplant patients have robust production of erythropoietin and normalization of hemoglobin, *anemia* is commonly seen in the posttransplant period. Often the anemia is attributable to bone marrow–suppressant immunosuppressive medications such as azathioprine, mycophenolic acid, and mTOR inhibitors. Gastrointestinal bleeding is a common side effect of high-dose and long-term steroid administration. Many transplant patients have creatinine clearances of 30–50 mL/min and can be considered to have chronic renal insufficiency for anemia management, including supplemental erythropoietin.

Chronic hepatitis, particularly when due to hepatitis B virus, can be a progressive, fatal disease over a decade or so. Patients who are persistently hepatitis B surface antigen–positive are at higher risk, according to some studies, but the presence of HCV is also a concern when one embarks on a course of immunosuppression in a transplant recipient. However, the introduction of the new highly effective, direct-acting HCV antiviral medications reduced this risk significantly.

In conclusion, while kidney transplantation has progressed significantly toward the goals of longer patient survival and better quality of life, the field still has significant challenges and unmet needs. Advanced immunologic and genetic studies have led and will continue to lead us to detailed understanding of alloimmunity at the molecular level. Noninvasive biomarkers for monitoring and diagnosing rejection and novel therapeutic targets will continue to evolve. Further effort is needed to achieve equity and improve personalized care of kidney transplant recipients.

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314 Glomerular Diseases

Julia B. Lewis, Eric G. Neilson

Two human kidneys harbor nearly 1.8 million glomerular capillary tufts. Each glomerular tuft resides within Bowman's space. The capsule circumscribing this space is lined by parietal epithelial cells that transition into tubular epithelia forming the proximal nephron or migrate into the tuft to replenish podocytes. The glomerular capillary tuft derives from an afferent arteriole that forms a branching capillary bed embedded in mesangial matrix **(Fig. 314-1)**. This capillary network funnels into an efferent arteriole, which passes filtered blood into cortical peritubular capillaries or medullary vasa recta that supply and exchange with a folded tubular architecture. Hence, the glomerular capillary tuft, fed and drained by arterioles, represents an arteriolar portal system. Fenestrated endothelial cells resting on a glomerular basement membrane (GBM) line glomerular capillaries. Delicate foot processes extending from epithelial podocytes shroud the outer surface of these capillaries, and adjacent podocytes interconnect to each other by slit-pore membranes forming a selective filtration barrier.

The glomerular capillaries filter 120–180 L/d of plasma water containing various solutes for reclamation or discharge by downstream tubules. Most large proteins and all cells are excluded from filtration by a physicochemical barrier governed by pore size and negative electrostatic charge. The mechanics of filtration and reclamation are quite complicated for many solutes **(Chap. 309)**. For example, in the case of serum albumin, the glomerulus is an imperfect barrier. Although albumin has a negative charge, which would tend to repel the negatively charged GBM, it only has a physical radius of 3.6 nm, while pores in the GBM and slit-pore membranes have a radius of 4 nm. Consequently, variable amounts of albumin inevitably cross the filtration barrier to be reclaimed by megalin and cubilin receptors along the proximal tubule. Remarkably, humans with normal nephrons excrete on average 8–10 mg of albumin in daily voided urine, ~20–60% of total excreted protein. This amount of albumin, and other proteins, can rise to gram quantities following glomerular injury.

The breadth of diseases affecting the glomerulus is expansive because the microenvironment supporting the glomerular capillaries can be injured in a variety of ways, producing many different lesions. Some order to this vast subject is brought by grouping all of these diseases into a smaller number of clinical syndromes.

PATHOGENESIS OF GLOMERULAR DISEASE

There are many forms of glomerular disease with pathogenesis variably linked to the presence of genetic mutations, infection, toxin exposure, autoimmunity, atherosclerosis, hypertension, emboli, thrombosis, or diabetes mellitus. Even after careful study, however, the cause often

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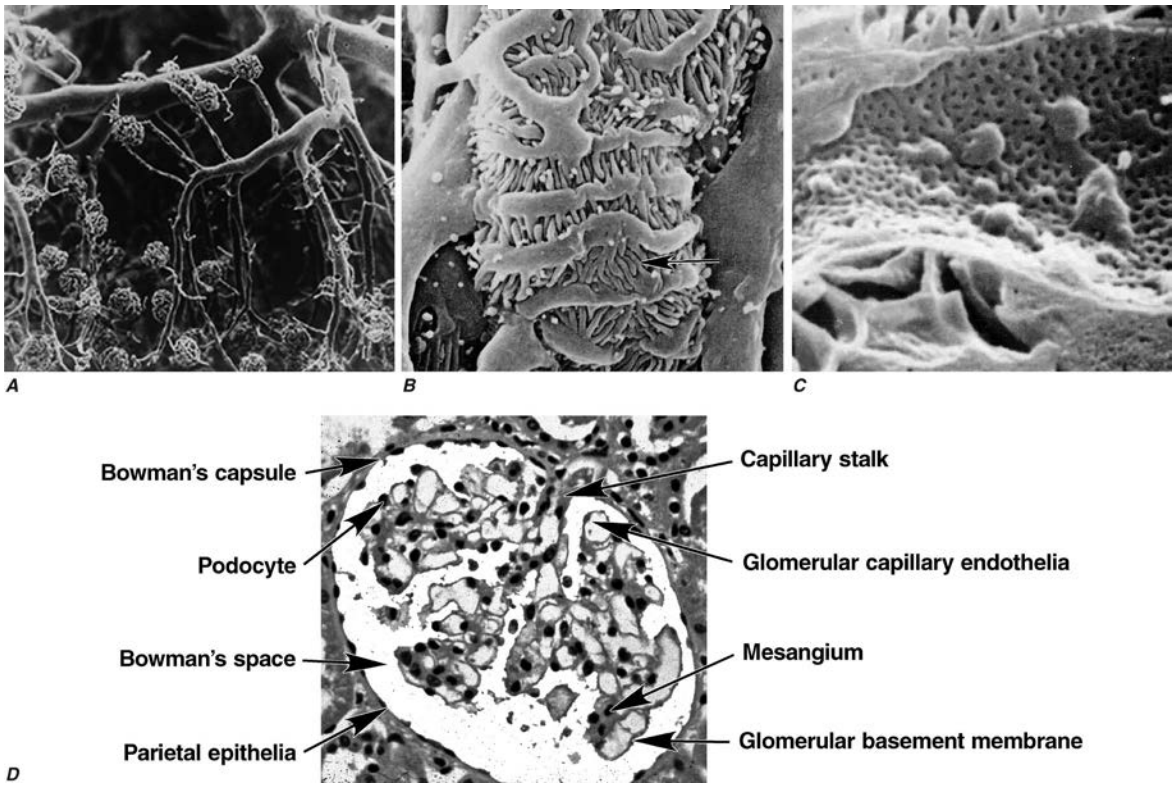


FIGURE 314-1 Glomerular architecture. ***A.*** The glomerular capillaries form from a branching network of renal arteries, arterioles leading to an afferent arteriole, glomerular capillary bed (tuft), and a draining efferent arteriole. *(From VH Gattone II et al: Hypertension 5:8, 1983.)* ***B.*** Scanning electron micrograph of podocytes that line the outer surface of the glomerular capillaries (*arrow* shows foot process). ***C.*** Scanning electron micrograph of the fenestrated endothelia lining the glomerular capillary. ***D.*** The various normal regions of the glomerulus on light microscopy. *(A–C: Courtesy of Dr. Vincent Gattone, Indiana University; with permission.)*

remains unknown, and the lesion is called *idiopathic*. Specific or unique features of pathogenesis are mentioned with the description of each of the glomerular diseases later in this chapter.

Some glomerular diseases result from genetic mutations producing familial disease or a founder effect: congenital nephrotic syndrome from mutations in *NPHS1* (nephrin) and *NPHS2* (podocin) affects the slit-pore membrane at birth, and *TRPC6* cation channel mutations produce *focal segmental glomerulosclerosis* (*FSGS*) in adulthood; polymorphisms in the gene encoding apolipoprotein L1, *APOL1*, are a major risk for nearly 70% of African Americans with nondiabetic end-stage renal disease (ESRD), particularly FSGS; monogenetic causes of FSGS are increasingly linked to early age of onset and to genes encoding type IV collagen in older adults, suggesting that much of FSGS may be hereditary; mutations in control of the complement pathway increasingly associate with various forms of *membranoproliferative glomerulonephritis* (*MPGN*) and C_3 glomerulopathies including dense deposit disease, or *atypical hemolytic-uremic syndrome* (*aHUS*); type II partial lipodystrophy from mutations in genes encoding lamin A/C or PPARγ cause a metabolic syndrome associated with MPGN; Alport's syndrome, from mutations in the genes encoding for the α3, α4, or α5 chains of type IV collagen, produces *split basement membranes* with *glomerulosclerosis*; and lysosomal storage diseases, such as α-galactosidase A deficiency causing Fabry's disease and *N*-acetylneuraminic acid hydrolase deficiency causing nephrosialidosis, produce FSGS.

Systemic hypertension and atherosclerosis can produce pressure stress, ischemia, or lipid oxidants that lead to *chronic glomerulosclerosis*. *Malignant hypertension* can quickly complicate glomerulosclerosis with fibrinoid necrosis of arterioles and glomeruli, thrombotic microangiopathy, and acute renal failure. *Diabetic nephropathy* is an acquired sclerotic injury associated with thickening of the GBM secondary to the long-standing effects of hyperglycemia, advanced glycosylation end products, and reactive oxygen species.

Inflammation of the glomerular capillaries is called *glomerulonephritis*. Most glomerular or mesangial antigens involved in *immune-mediated glomerulonephritis* are unknown **(Fig. 314-2)**. Glomerular epithelial or mesangial cells may shed or express epitopes that mimic other immunogenic proteins made elsewhere in the body. Bacteria, fungi, and viruses can directly infect the kidney producing their own antigens. Autoimmune diseases such as idiopathic *membranous glomerulonephritis* (*MGN*) or MPGN are confined to the kidney, whereas systemic inflammatory diseases such as *lupus nephritis* or *granulomatosis with polyangiitis* spread to the kidney, causing secondary glomerular injury. *Antiglomerular basement membrane disease* producing Goodpasture's syndrome primarily injures both the lung and kidney because of the narrow distribution of the α3 NC1 domain of type IV collagen that is the target antigen.

Local activation of Toll-like receptors on glomerular cells, deposition of immune complexes, or complement injury to glomerular structures induces mononuclear cell infiltration, which subsequently leads to an adaptive immune response attracted to the kidney by local release of chemokines. Neutrophils, macrophages, and T cells are drawn by chemokines into the glomerular tuft, where they react with antigens and epitopes on or near somatic cells or their structures, producing more cytokines and proteases that damage the mesangium, capillaries, and/or the GBM. While the adaptive immune response is similar to that of other tissues, early T-cell activation plays an important role in the mechanism of glomerulonephritis. Antigens presented by class II major histocompatibility complex (MHC) molecules on macrophages and dendritic cells in conjunction with associative recognition molecules engage the CD4/8 T-cell repertoire.

Mononuclear cells by themselves can injure the kidney, but autoimmune events that damage glomeruli classically produce a humoral immune response. *Poststreptococcal glomerulonephritis, lupus nephritis*, and idiopathic *membranous nephritis* typically are associated with immune deposits along the GBM, while anti-GBM antibodies produce the linear binding of anti-GBM disease. Preformed circulating immune complexes can precipitate along the subendothelial side of the GBM, while other immune deposits form in situ on the subepithelial side. These latter deposits accumulate when circulating autoantibodies find their antigen trapped along the subepithelial edge of the GBM.

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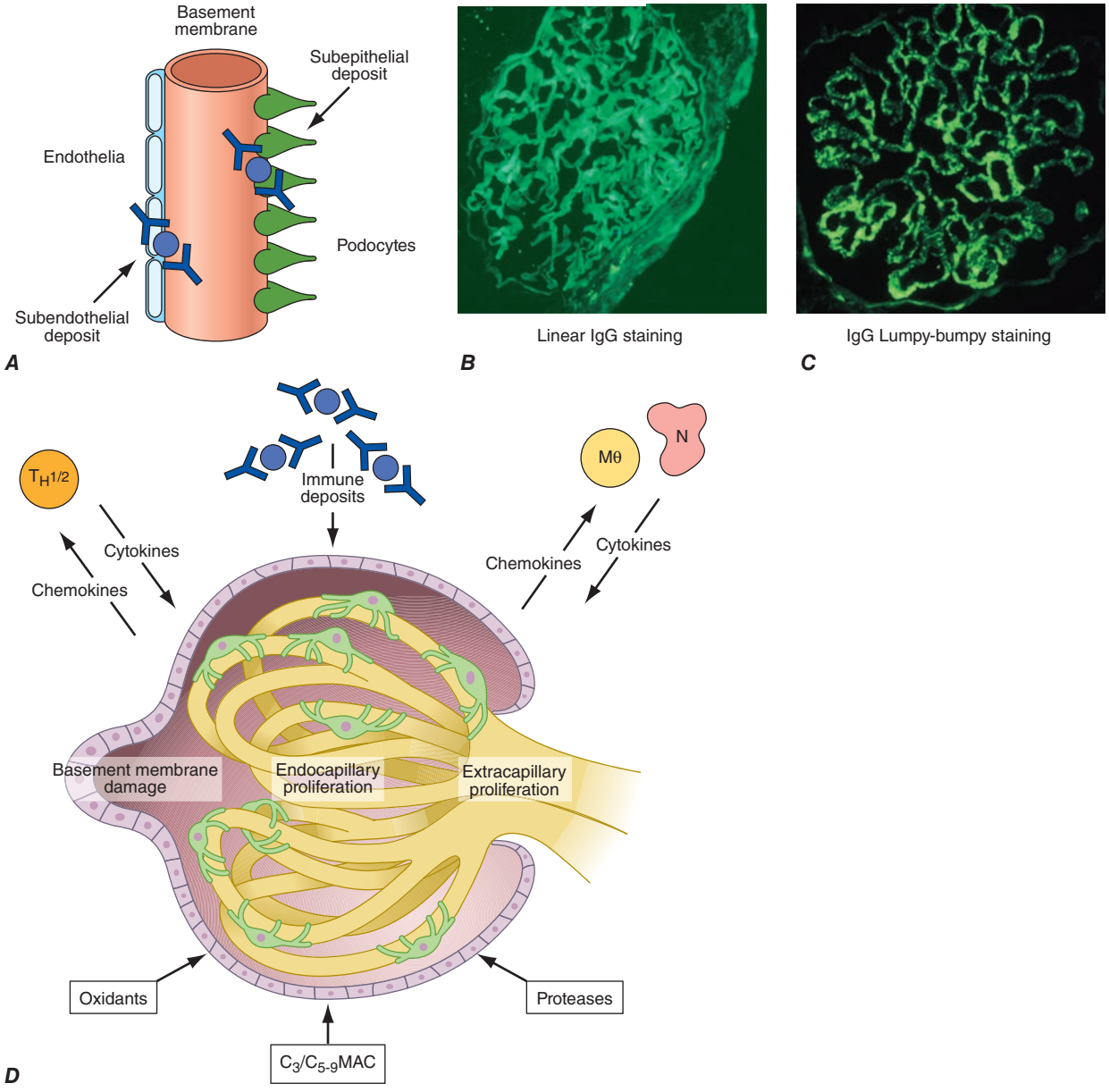


FIGURE 314-2 The glomerulus is injured by a variety of mechanisms. ***A.*** Preformed immune deposits can precipitate from the circulation and collect along the glomerular basement membrane (GBM) in the subendothelial space or can form in situ along the subepithelial space. ***B.*** Immunofluorescent staining of glomeruli with labeled anti-IgG demonstrating linear staining from a patient with anti-GBM disease or immune deposits from a patient with membranous glomerulonephritis. ***C.*** The mechanisms of glomerular injury have a complicated pathogenesis. Immune deposits and complement deposition classically draw macrophages and neutrophils into the glomerulus. T lymphocytes may follow to participate in the injury pattern as well. ***D.*** Amplification mediators as locally derived oxidants and proteases expand this inflammation, and depending on the location of the target antigen and the genetic polymorphisms of the host, basement membranes are damaged with either endocapillary or extracapillary proliferation.

Immune deposits in the glomerular mesangium may result from the deposition of preformed circulating complexes or in situ antigen-antibody interactions. Immune deposits stimulate the release of local proteases and activate the complement cascade, producing C_{5-9} attack complexes. In addition, local oxidants damage glomerular structures, producing proteinuria and effacement of the podocytes. Overlapping etiologies or pathophysiologic mechanisms can produce similar glomerular lesions, suggesting that downstream molecular and cellular responses often converge toward common patterns of injury.

PROGRESSION OF GLOMERULAR DISEASE

Persistent glomerulonephritis that worsens renal function is always accompanied by interstitial nephritis, renal fibrosis, and tubular atrophy (see Fig. A4-27). What is not so obvious, however, is that renal failure in glomerulonephritis best correlates histologically with the appearance of tubulointerstitial nephritis rather than with the type of inciting glomerular injury.

Loss of renal function due to interstitial damage is explained hypothetically by several mechanisms. The simplest explanation is that urine flow is impeded by tubular obstruction as a result of interstitial inflammation and fibrosis. Thus, obstruction of the tubules with debris or by extrinsic compression functionally results in aglomerular nephrons. A second mechanism suggests that interstitial changes, including interstitial edema or fibrosis, alter tubular and vascular architecture and thereby compromise the normal tubular transport of solutes and water from tubular lumen to vascular space. This failure increases the solute and water content of the tubule fluid, resulting in isosthenuria and polyuria. Adaptive mechanisms related to tubuloglomerular feedback also fail, resulting in a reduction of renin output from the juxtaglomerular apparatus trapped by interstitial inflammation. Consequently, the local vasoconstrictive influence of angiotensin II on the glomerular arterioles decreases, and filtration drops owing to a generalized decrease in arteriolar tone. A third mechanism involves changes in vascular resistance due to damage of peritubular capillaries.

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The cross-sectional volume of these capillaries is decreased by interstitial inflammation, edema, or fibrosis. These structural alterations in vascular resistance affect renal function through two mechanisms. First, tubular cells are very metabolically active, and as a result, decreased perfusion leads to tubular ischemic injury. Second, impairment of glomerular arteriolar outflow leads to increased intravascular hypertension in less-involved glomeruli; this selective intraglomerular hypertension aggravates and extends *mesangial sclerosis* and *glomerulosclerosis* to less-involved glomeruli. Regardless of the exact mechanism, early *acute tubulointerstitial nephritis* **(see Fig. A4-27)** suggests potentially recoverable renal function, whereas the development of *chronic interstitial fibrosis* prognosticates permanent loss **(see Fig. A4-30)**.

Persistent damage to glomerular capillaries spreads to the tubulointerstitium in association with proteinuria. There is a hypothesis that efferent arterioles leading from inflamed glomeruli carry forward inflammatory mediators, which induces downstream interstitial nephritis, resulting in fibrosis. Glomerular filtrate from injured glomerular capillaries adherent to Bowman's capsule may also be misdirected to the periglomerular interstitium. Most nephrologists believe, however, that proteinuric glomerular filtrate forming tubular fluid is the primary route to downstream tubulointerstitial injury, although none of these hypotheses are mutually exclusive.

The simplest explanation for the effect of proteinuria on the development of interstitial nephritis is that increasingly severe proteinuria, carrying activated cytokines and lipoproteins producing reactive oxygen species, triggers a downstream inflammatory cascade in and around epithelial cells lining the tubular nephron. These effects induce T lymphocyte and macrophage infiltrates in the interstitial spaces along with fibrosis and tubular atrophy.

Tubules disaggregate following direct damage to their basement membranes, leading to more interstitial fibroblasts and fibrosis at the site of injury; recent comprehensive evidence suggests that renal fibroblasts increase through several mechanisms: epithelial or endothelial-mesenchymal transitions (15%), bone marrow–derived fibrocytes (35%), and the proliferation of resident fibroblasts (50%). Transforming growth factor β (TGF-β), fibroblast growth factor 2 (FGF-2), hypoxemia-inducible factor 1α (HIF-1α), and platelet-derived growth factor (PDGF) are particularly active in this transition. With persistent nephritis, fibroblasts multiply and lay down tenascin and a fibronectin scaffold for the polymerization of new interstitial collagen types I/III. These events form scar tissue through a process called fibrogenesis. In experimental studies, bone morphogenetic protein 7 and hepatocyte growth factor can reverse early fibrogenesis and preserve tubular architecture. When fibroblasts outdistance their survival factors, apoptosis occurs, and the permanent renal scar becomes acellular, leading to irreversible renal failure.

APPROACH TO THE PATIENT

Glomerular Disease

HEMATURIA, PROTEINURIA, AND PYURIA

Patients with glomerular disease usually have some hematuria with varying degrees of proteinuria. Hematuria is typically asymptomatic. As few as 3–5 red blood cells in the spun sediment from first-voided morning urine is suspicious. The diagnosis of glomerular injury can be delayed because patients will not realize they have *microscopic hematuria*, and only rarely with the exception of IgA nephropathy and sickle cell disease is *gross hematuria* present. When working up microscopic hematuria, perhaps accompanied by minimal proteinuria (<500 mg/24 h), it is important to exclude anatomic lesions, such as malignancy of the urinary tract, particularly in older men. Microscopic hematuria may also appear with the onset of benign prostatic hypertrophy, interstitial nephritis, papillary necrosis, hypercalciuria, renal stones, cystic kidney diseases, or renal vascular injury. However, when red blood cell casts **(see Fig. A4-34)** or dysmorphic red blood cells are found in the sediment, glomerulonephritis is likely. A mean of 8–10 mg/24 h of albumin appears in the urine in the absence of kidney disease. In early nephropathy, such as in diabetic nephropathy, proteinuria increases to 30–300 mg/24 h and is called microalbuminuria and represents the presence of renal disease. Greater than 300 mg/24 h of albuminuria represents frank proteinuria and more advanced renal disease **(Table 314-1)**.

Sustained proteinuria >1–2 g/24 h is also commonly associated with glomerular disease. Patients often will not know they have proteinuria unless they become edematous or notice foaming urine on voiding. *Sustained proteinuria* has to be distinguished from lesser amounts of so-called *benign proteinuria* in the normal population. (Table 314-1). This latter class of proteinuria is nonsustained, generally <1 g/24 h, and is sometimes called *functional* or *transient proteinuria*. Fever, exercise, obesity, sleep apnea, emotional stress, and congestive heart failure can explain transient proteinuria. Proteinuria only seen with upright posture is called *orthostatic proteinuria* and has a benign prognosis. Isolated proteinuria sustained over multiple clinic visits is found in many glomerular lesions. Proteinuria in most adults with glomerular disease is *nonselective*, containing albumin and a mixture of other serum proteins, whereas in children with *minimal change disease* (MCD), the proteinuria is *selective* and composed largely of albumin.

Some patients with inflammatory glomerular disease, such as acute poststreptococcal glomerulonephritis or MPGN, have *pyuria* characterized by the presence of considerable numbers of leukocytes. This latter finding has to be distinguished from urine infected with bacteria.

CLINICAL SYNDROMES

Various forms of glomerular injury can also be parsed into several distinct syndromes on clinical grounds **(Table 314-2)**. These syndromes, however, are not always mutually exclusive. There is an *acute nephritic syndrome* producing 1–2 g/24 h of proteinuria, hematuria with red blood cell casts, pyuria, hypertension, fluid retention, and a rise in serum creatinine associated with a reduction in glomerular filtration. If glomerular inflammation develops slowly, the serum creatinine will rise gradually over many weeks, but if the serum creatinine rises quickly, particularly over a few days, acute nephritis is sometimes called *rapidly progressive glomerulonephritis* (RPGN); the histopathologic term *crescentic glomerulonephritis* is the pathologic equivalent of the clinical presentation of RPGN. When patients with RPGN present with lung hemorrhage from Goodpasture's syndrome, antineutrophil cytoplasmic antibody (ANCA)–associated small-vessel vasculitis, lupus erythematosus, or cryoglobulinemia, they are often diagnosed as having a *pulmonary-renal syndrome*. *Nephrotic syndrome* describes the onset of heavy proteinuria (>3.0 g/24 h), hypertension, hypercholesterolemia, hypoalbuminemia, edema/anasarca,

TABLE 314-1 Urine Assays for Albuminuria/Proteinuria

	24-h ALBUMIN[a] (mg/24 h)	ALBUMIN[a]/CREATININE RATIO (mg/g)	DIPSTICK PROTEINURIA	24-H URINE PROTEIN[b] (mg/24 h)
Normal	8–10	<30	–	<150
Microalbuminuria	30–300	30–300	–/Trace/1+	–/>150
Proteinuria	>300	>300	Trace–3+	>150

[a]Albumin detected by radioimmunoassay. [b]Albumin represents 20–60% of the total protein excreted in the urine.

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TABLE 314-2 Patterns of Clinical Glomerulonephritis

GLOMERULAR SYNDROMES	PROTEINURIA	HEMATURIA	VASCULAR INJURY
Acute Nephritic Syndromes			
Poststreptococcal glomerulonephritis[a]	+/++	++/+++	–
Subacute bacterial endocarditis[a]	+/++	++	–
Lupus nephritis[a]	+/++	++/+++	+
Antiglomerular basement membrane disease[a]	++	++/+++	–
IgA nephropathy[a]	+/++	+++[c]	–
ANCA small-vessel vasculitis[a]			
Granulomatosis with polyangiitis (Wegener's)	+/++	++/+++	++++
Microscopic polyangiitis	+/++	++/+++	++++
Churg-Strauss syndrome	+/++	++/+++	++++
Henoch-Schönlein purpura[a]	+/++	++/+++[c]	++++
Cryoglobulinemia[a]	+/++	++/+++	++++
Membranoproliferative glomerulonephritis[a]	++	++/+++	–
C_3 glomerulopathies	++	++/+++	-
Mesangioproliferative glomerulonephritis	+	+/++	–
Pulmonary-Renal Syndromes			
Goodpasture's syndrome[a]	++	++/+++	–
ANCA small-vessel vasculitis[a]			
Granulomatosis with polyangiitis (Wegener's)	+/++	++/+++	++++
Microscopic polyangiitis	+/++	++/+++	++++
Churg-Strauss syndrome	+/++	++/+++	++++
Henoch-Schönlein purpura[a]	+/++	++/+++[c]	++++
Cryoglobulinemia[a]	+/++	++/+++	++++
Nephrotic Syndromes			
Minimal change disease	++++	–	–
Focal segmental glomerulosclerosis	+++/++++	+	–
Membranous glomerulonephritis	++++	+	–
Diabetic nephropathy	++/++++	–/+	–
AL and AA amyloidosis	+++/++++	+	+/++
Light chain deposition disease	+++	+	–
Fibrillary-immunotactoid disease	+++/++++	+	+
Fabry's disease	+	+	–
Basement Membrane Syndromes			
Anti-GBM disease[a]	++	++/+++	–
Alport's syndrome	++	++	–
Thin basement membrane disease	+	++	–
Nail-patella syndrome	++/+++	++	–
Glomerular Vascular Syndromes			
Atherosclerotic nephropathy	+	+	+++
Hypertensive nephropathy[b]	+/++	+/++	++
Cholesterol emboli	+/++	++	+++
Sickle cell disease	+/++	+++[c]	+++
Thrombotic microangiopathies	++	++	+++
Antiphospholipid syndrome	++	++	+++
ANCA small-vessel vasculitis[a]			
Granulomatosis with polyangiitis (Wegener's)	+/++	++/+++	++++
Microscopic polyangiitis	+/++	++/+++	++++
Churg-Strauss syndrome	+++	++/+++	++++
Henoch-Schönlein purpura[a]	+/++	++/+++[c]	++++
Cryoglobulinemia[a]	+/++	++/+++	++++
AL and AA amyloidosis	+++/++++	+	+/++
Infectious Disease–Associated Syndromes			
Poststreptococcal glomerulonephritis[a]	+/++	++/+++	–
Subacute bacterial endocarditis[a]	+/++	++	–
HIV	+++	+/++	–

(*Continued*)

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TABLE 314-2 Patterns of Clinical Glomerulonephritis (Continued)

GLOMERULAR SYNDROMES	PROTEINURIA	HEMATURIA	VASCULAR INJURY
Hepatitis B and C	+++	+/++	–
Syphilis	+++	+	–
Leprosy	+++	+	–
Malaria	+++	+/++	–
Schistosomiasis	+++	+/++	–

[a]Can present as rapidly progressive glomerulonephritis (RPGN); sometimes called crescentic glomerulonephritis. [b]Can present as a malignant hypertensive crisis producing an aggressive fibrinoid necrosis in arterioles and small arteries with microangiopathic hemolytic anemia. [c]Can present with gross hematuria.

Abbreviations: AA, amyloid A; AL, amyloid L; ANCA, antineutrophil cytoplasmic antibodies; GBM, glomerular basement membrane.

and microscopic hematuria; if only large amounts of proteinuria are present without clinical manifestations, the condition is sometimes called *nephrotic-range proteinuria*. The glomerular filtration rate (GFR) in these patients may initially be normal or, rarely, higher than normal, but with persistent hyperfiltration and continued nephron loss, it typically declines over months to years. Patients with a *basement membrane syndrome* either have genetically abnormal basement membranes (Alport's syndrome) or an autoimmune response to basement membrane collagen IV (Goodpasture's syndrome) associated with microscopic hematuria, mild to heavy proteinuria, and hypertension with variable elevations in serum creatinine. *Glomerular-vascular syndrome* describes patients with vascular injury producing hematuria and moderate proteinuria. Affected individuals can have vasculitis, thrombotic microangiopathy, antiphospholipid syndrome, or, more commonly, a systemic disease such as atherosclerosis, cholesterol emboli, hypertension, sickle cell anemia, and autoimmunity. *Infectious disease–associated syndrome* is most important if one has a global perspective. Save for subacute bacterial endocarditis (SBE) in the Western Hemisphere, malaria and schistosomiasis may be the most common causes of glomerulonephritis throughout the world, closely followed by HIV and chronic hepatitis B and C. These infectious diseases produce a variety of inflammatory reactions in glomerular capillaries, ranging from nephrotic syndrome to acute nephritic injury, and urinalyses that demonstrate a combination of hematuria and proteinuria.

These six general categories of syndromes are usually determined at the bedside with the help of a history and physical examination, blood chemistries, renal ultrasound, and urinalysis. These initial studies help frame further diagnostic workup that typically involves testing of the serum for the presence of various proteins (HIV and hepatitis B and C antigens) or antibodies (anti-GBM, antiphospholipid, antistreptolysin O [ASO], PLA2R, THSD7A, anti-DNAse, antihyaluronidase, ANCA, anti-DNA, cryoglobulins, anti-HIV, and anti-hepatitis B and C antibodies) or depletion of complement components (C_3 and C_4). The bedside history and physical examination can also help determine whether the glomerulonephritis is isolated to the kidney (*primary glomerulonephritis*) or is part of a systemic disease (*secondary glomerulonephritis*).

When confronted with an abnormal urinalysis and elevated serum creatinine, with or without edema or congestive heart failure, one must consider whether the glomerulonephritis is *acute* or *chronic*. This assessment is best made by careful history (last known urinalysis or serum creatinine during pregnancy or insurance physical, evidence of infection, or use of medication or recreational drugs), the size of the kidneys on renal ultrasound examination, and how the patient feels at presentation. Chronic glomerular disease often presents with decreased kidney size. Patients who quickly develop renal failure are fatigued and weak and often have uremic symptoms associated with nausea, vomiting, fluid retention, and somnolence. Primary glomerulonephritis presenting with renal failure that has progressed slowly, however, can be remarkably asymptomatic, as are patients with acute glomerulonephritis without much loss in renal function. Once this initial information is collected, selected patients who are clinically stable, have adequate blood clotting parameters, and are willing and able to receive treatment are encouraged to have a renal biopsy.

RENAL PATHOLOGY

A renal biopsy in the setting of glomerulonephritis quickly identifies the type of glomerular injury and often suggests a course of treatment. The biopsy is processed for light microscopy using stains for *hematoxylin and eosin* (*H&E*) to assess cellularity and architecture, *periodic acid–Schiff* (*PAS*) to stain carbohydrate moieties in the membranes of the glomerular tuft and tubules, *Jones-methenamine silver* to enhance basement membrane structure, *Congo red* for amyloid deposits, and *Masson's trichrome* to identify collagen deposition and assess the degree of glomerulosclerosis and interstitial fibrosis. Biopsies are also processed for direct immunofluorescence using conjugated antibodies against IgG, IgM, and IgA to detect the presence of "lumpy-bumpy" immune deposits or "linear" IgG or IgA antibodies bound to GBM, antibodies against trapped complement proteins (C_3 and C_4), or specific antibodies against a relevant antigen (PLA2R, THSD7A, and DNAJB9). High-resolution electron microscopy can clarify the principal location of immune deposits and the status of the basement membrane.

Each region of a renal biopsy is assessed separately. By light microscopy, glomeruli (ideally 20) are reviewed individually for discrete lesions; <50% involvement is considered *focal*, and >50% is *diffuse*. Injury in each glomerular tuft can be *segmental,* involving a portion of the tuft, or *global,* involving most of the glomerulus. Glomeruli having *proliferative* characteristics show increased cellularity. When cells in the capillary tuft proliferate, it is called *endocapillary*, and when cellular proliferation extends into Bowman's space, it is called *extracapillary. Synechiae* are formed when epithelial podocytes attach to Bowman's capsule in the setting of glomerular injury; *crescents*, which in some cases may be the extension of synechiae, develop when fibrocellular/fibrin collections fill all or part of Bowman's space; and *sclerotic* glomeruli show acellular, amorphous accumulations of proteinaceous material throughout the tuft with loss of functional capillaries and normal mesangium. Since *age-related glomerulosclerosis* is common in adults, one can estimate the background percentage of sclerosis by dividing the patient's age in half and subtracting 10. Immunofluorescent and electron microscopy can detect the presence and location of *subepithelial, subendothelial, or mesangial* immune deposits, or *reduplication* or *splitting* of the basement membrane. In the other regions of the biopsy, the vasculature surrounding glomeruli and tubules can show *angiopathy, vasculitis*, the presence of *fibrils*, or *thrombi*. The tubules can be assessed for adjacency to one another; separation can be the result of edema, tubular dropout, or collagen deposition resulting from interstitial fibrosis. Interstitial fibrosis is an ominous sign of irreversibility and progression to renal failure.

ACUTE NEPHRITIC SYNDROMES

Acute nephritic syndromes classically present with hypertension, hematuria, red blood cell casts, pyuria, and mild to moderate proteinuria. Extensive inflammatory damage to glomeruli causes a fall in GFR and eventually produces uremic symptoms with salt and water retention, leading to edema and hypertension.

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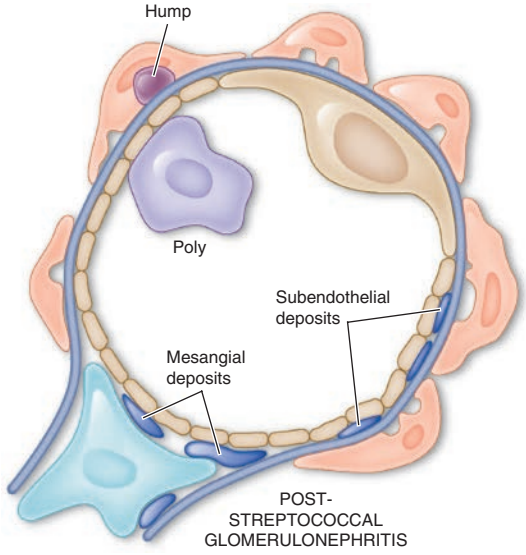
POSTSTREPTOCOCCAL GLOMERULONEPHRITIS

Poststreptococcal glomerulonephritis is prototypical for *acute endocapillary proliferative glomerulonephritis*. The incidence of poststreptococcal glomerulonephritis has dramatically decreased in developed countries, and in these locations is typically sporadic. Acute nephritis in underdeveloped countries is epidemic and usually affects children between the ages of 2 and 14 years. In developed countries, it is more typical in the elderly, especially in association with debilitating conditions. It is more common in males, and the familial or cohabitant incidence is as high as 40%. Skin and more commonly throat infections with particular M types of streptococci (nephritogenic strains) antedate glomerular disease. Antibiotic therapy does not reduce the occurrence of nephritis. Poststreptococcal glomerulonephritis due to pharyngitis develops 1–3 weeks after infection and 2–6 weeks after impetigo.

The renal biopsy in poststreptococcal glomerulonephritis demonstrates hypercellularity of mesangial and endothelial cells; glomerular infiltrates of polymorphonuclear leukocytes; granular subendothelial immune deposits of IgG, IgM, C_3, C_4, and C_{5-9}; and subepithelial deposits (which appear as "humps") **(see Fig. A4-6)**. **(See Glomerular Schematic 1.)** Poststreptococcal glomerulonephritis is an immune-mediated disease involving putative streptococcal antigens, circulating immune complexes, and activation of complement in association with cell-mediated injury. Many candidate antigens have been proposed over the years; candidates from nephritogenic streptococci are a cationic cysteine proteinase known as streptococcal pyrogenic exotoxin B (SPEB) and NAPlr, the nephritis-associated plasmin receptor. The nephritogenic antigen SPEB has been demonstrated inside the subepithelial "humps" on biopsy.

The classic presentation is an acute nephritic picture with hematuria, pyuria, red blood cell casts, edema, hypertension, and oliguric renal failure, which may be severe enough to appear as RPGN. Systemic symptoms of headache, malaise, anorexia, and flank pain (due to swelling of the renal capsule) are reported in as many as 50% of cases. Five percent of children and 20% of adults have proteinuria in the nephrotic range. In the first week of symptoms, 90% of patients will have a depressed CH_{50} and decreased levels of C_3 with normal levels of C_4. Positive rheumatoid factor (30–40%), cryoglobulins, circulating immune complexes (60–70%), and ANCA against myeloperoxidase (10%) are also reported. Positive cultures for streptococcal infection are inconsistently present (10–70%), but increased titers of ASO (30%), anti-DNAse (70%), or antihyaluronidase antibodies (40%) can help confirm the diagnosis. Consequently, the diagnosis of poststreptococcal glomerulonephritis rarely requires a renal biopsy. A subclinical disease is reported in some series to be 4–5 times as common as clinical nephritis, and these latter cases are characterized by asymptomatic microscopic hematuria with low serum C_3 complement levels.

Glomerular schematic 1



Treatment is supportive, with control of hypertension, edema, and dialysis as needed. Antibiotic treatment for active streptococcal infection should be given to patients and their cohabitants. There is no role for immunosuppressive therapy, even in the setting of crescents. Recurrent poststreptococcal glomerulonephritis is rare despite repeated streptococcal infections. Early death is rare in children but does occur in the elderly. Complete resolution of the azotemia, hematuria, and proteinuria in the majority of children occurs within 3–6 weeks of the onset of nephritis, but 3–10% of children may have persistent microscopic hematuria, nonnephrotic proteinuria, or hypertension. Overall, the prognosis is good, with ESRD being very uncommon in children and adults. The prognosis in elderly patients is worse, with a high incidence of azotemia (up to 60%), nephrotic-range proteinuria, and ESRD.

SUBACUTE BACTERIAL ENDOCARDITIS

Endocarditis-associated glomerulonephritis is typically a complication of SBE, particularly in patients who remain untreated for a long time, have negative blood cultures, or have right-sided endocarditis. Common comorbidities are valvular heart disease, intravenous drug use, hepatitis C, and diabetes mellitus. Glomerulonephritis is unusual in acute bacterial endocarditis because it takes 10–14 days to develop immune complex–mediated injury, by which time the patient has been treated, often with emergent surgery. Grossly, the kidneys in SBE have subcapsular hemorrhages with a "flea-bitten" appearance, and microscopy on renal biopsy reveals focal proliferation around foci of necrosis associated with abundant mesangial, subendothelial, and subepithelial immune deposits of IgG, IgM, and C_3. Commonly patients present with a clinical picture of RPGN and have crescents on biopsy. Embolic infarcts or septic abscesses may also be present. The pathogenesis hinges on the renal deposition of circulating immune complexes in the kidney with complement activation. Patients present with gross or microscopic hematuria, pyuria, and mild proteinuria, acute kidney injury, or RPGN with rapid loss of renal function. A normocytic anemia, elevated erythrocyte sedimentation rate, hypocomplementemia, high titers of rheumatoid factor, type III cryoglobulins, circulating immune complexes, and ANCAs may be present. Levels of serum creatinine may be elevated at diagnosis, but with modern therapy, there is little progression to chronic renal failure. Primary treatment is eradication of the infection with 4–6 weeks of antibiotics, and if accomplished expeditiously, the prognosis for renal recovery is good. ANCA-associated vasculitis sometimes accompanies or is confused with SBE and should be ruled out, as the treatment is different.

As variants of persistent bacterial infection in blood-associated glomerulonephritis, postinfectious glomerulonephritis can occur in patients with ventriculoatrial and ventriculoperitoneal shunts; pulmonary, intraabdominal, pelvic, or cutaneous infections; and infected vascular prostheses. In developed countries, a significant proportion of cases afflict adults, especially the immunocompromised, and the predominant organism is *Staphylococcus*. The clinical presentation of these conditions is variable and includes proteinuria, microscopic hematuria, acute renal failure, and hypertension. Serum complement levels are low, and there may be elevated levels of C-reactive proteins, rheumatoid factor, antinuclear antibodies, and cryoglobulins. Renal lesions include MPGN, diffuse proliferative and exudative glomerulonephritis (DPGN), or mesangioproliferative glomerulonephritis, sometimes leading to RPGN. Treatment focuses on eradicating the infection, with most patients treated as if they have endocarditis. The prognosis is guarded.

LUPUS NEPHRITIS

Lupus nephritis is a common and serious complication of systemic lupus erythematosus (SLE). Clinical manifestations of renal disease are present in 30% of patients at the time of diagnosis, and the majority will develop renal abnormalities in the course of their disease; it is more common in blacks, Asians, and Hispanics than it is in whites. Lupus nephritis results from the deposition of circulating immune complexes composed of primarily DNA and anti-DNA, which activate the complement cascade, leading to complement-mediated damage,

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leukocyte infiltration, activation of procoagulant factors, and release of various cytokines. In situ immune complex formation also plays a role in renal injury. These immune deposits may occur in the mesangial, subendothelial, and/or subepithelial spaces.

The clinical manifestations, course of disease, and treatment of lupus nephritis are closely linked to renal pathology. The most common clinical sign of renal disease is proteinuria, but hematuria, hypertension, varying degrees of renal failure, and active urine sediment with red blood cell casts can all be present. Anti-dsDNA antibodies that fix complement correlate best with the presence of renal disease. Hypocomplementemia is common in patients with acute lupus nephritis (70–90%), and declining complement levels may herald a flare. A kidney biopsy should be performed in most patients with renal involvement to establish the histologic subtype, which guides therapy.

The World Health Organization (WHO) workshop in 1974 first outlined several distinct patterns of lupus-related glomerular injury, and this classification was modified in 2004. This latest version of lesions seen on biopsy **(Table 314-3)** forms the basis for modern treatment recommendations. Class I nephritis describes normal glomerular histology by normal light microscopy with minimal mesangial deposits on immunofluorescent or electron microscopy. Class II designates mesangial immune complexes with *mesangial proliferation*. Both class I and II lesions are typically associated with minimal renal manifestation and normal renal function; nephrotic syndrome is rare. Patients with lesions limited to the renal mesangium have an excellent prognosis and generally do not need therapy for their lupus nephritis.

The subject of lupus nephritis is presented under acute nephritic syndromes because of the aggressive and important proliferative lesions seen in class III–V renal diseases. Class III describes *focal lesions involving <50% of the glomeruli with proliferation or scarring*, often involving only a segment of the glomerulus **(see Fig. A4-12)**. Class III lesions have the most varied course. Hematuria and proteinuria are present, and some patients also have an active urinary sediment, nephrotic syndrome, hypertension, and a decreased GFR. Patients with mild proliferation involving a small percentage of glomeruli respond well to therapy with steroids alone, and <5% progress to renal failure over 5 years. Patients with more severe proliferation involving a greater percentage of glomeruli have a far worse prognosis and lower remission rates. Treatment of those patients is the same as that for class IV lesions. Class IV describes *diffuse lesions with >50% of the glomeruli involved and proliferative endocapillary lesions with or without extracapillary lesions that may be segmental (IV-S), involving <50% of the glomerular tuft, or global (IV-G), involving >50%*. Patients with class IV lesions commonly have high anti-DNA antibody titers, low serum complement, hematuria, red blood cell casts, proteinuria, hypertension, and decreased renal function; 50% of patients have nephrotic-range proteinuria. Patients with crescents on biopsy often have a rapidly progressive decline in renal function **(see Fig. A4-12)**. Without treatment, this aggressive lesion has the worst renal prognosis, with class IV-S worse than class IV-G. However, if a remission—defined as a return to near-normal renal function and proteinuria ≤330 mg/dL per day—is achieved with treatment, renal outcomes are excellent. Current evidence suggests that inducing a remission with administration of high-dose steroids and either cyclophosphamide or mycophenolate mofetil for 2–6 months, followed by maintenance therapy with lower doses of steroids and mycophenolate mofetil or azathioprine, best balances the likelihood of successful remission with the side effects of therapy. There is no consensus on use of high-dose intravenous methylprednisolone versus oral prednisone, monthly intravenous cyclophosphamide versus daily oral cyclophosphamide, or other immunosuppressants such as cyclosporine, tacrolimus, or rituximab. Nephrologists tend to avoid prolonged use of cyclophosphamide in patients of childbearing age without first banking eggs or sperm.

The class V lesion describes subepithelial immune deposits producing a *membranous pattern*; a subcategory of class V lesions is associated with proliferative lesions and is sometimes called *mixed membranous and proliferative disease* **(see Fig. A4-11)**; this category of injury is treated like class IV glomerulonephritis. Sixty percent of patients present with nephrotic syndrome or lesser amounts of proteinuria. Patients with lupus nephritis class V, like patients with *idiopathic membranous nephropathy* (IMN), are predisposed to renal vein thrombosis and other thrombotic complications. A minority of patients with class V will present with hypertension and renal dysfunction. There are conflicting data on the clinical course, prognosis, and appropriate therapy for patients with class V disease, which may reflect the heterogeneity of this group of patients. Patients with severe nephrotic syndrome, elevated serum creatinine, and a progressive course will probably benefit from therapy with steroids in combination with other immunosuppressive agents. Therapy with inhibitors of the renin-angiotensin system also may attenuate the proteinuria. Antiphospholipid antibodies present in lupus may result in glomerular microthromboses and a thrombotic microangiopathy. The renal prognosis is worse despite anticoagulant therapy.

Patients with any of the above lesions also can transform to another lesion; hence, patients often require reevaluation, including repeat renal biopsy. Lupus patients with class VI lesions have >90% *sclerotic glomeruli* and ESRD with interstitial fibrosis. Up to 20% of patients with lupus nephritis will reach end-stage disease, requiring dialysis or transplantation. Patients with lupus nephritis have a markedly increased mortality compared with the general population. Renal transplantation in renal failure from lupus, usually performed after ~6 months of inactive disease, results in allograft survival rates comparable to patients transplanted for other reasons.

TABLE 314-3 Classification for Lupus Nephritis

Class I	Minimal mesangial	Normal histology with mesangial deposits
Class II	Mesangial proliferation	Mesangial hypercellularity with expansion of the mesangial matrix
Class III	Focal nephritis	Focal endocapillary ± extracapillary proliferation with focal subendothelial immune deposits and mild mesangial expansion
Class IV	Diffuse nephritis	Diffuse endocapillary ± extracapillary proliferation with diffuse subendothelial immune deposits and mesangial alterations
Class V	Membranous nephritis	Thickened basement membranes with diffuse subepithelial immune deposits; may occur with class III or IV lesions and is sometimes called mixed membranous and proliferative nephritis
Class VI	Sclerotic nephritis	Global sclerosis of nearly all glomerular capillaries

Note: Revised in 2004 by the International Society of Nephrology-Renal Pathology Society Study Group.

ANTIGLOMERULAR BASEMENT MEMBRANE DISEASE

Patients who develop autoantibodies directed against glomerular basement antigens frequently develop a glomerulonephritis termed *antiglomerular basement membrane* (*anti-GBM*) *disease*. When they present with lung hemorrhage and glomerulonephritis, they have a pulmonary-renal syndrome called *Goodpasture's syndrome*. The target epitopes for this autoimmune disease lie in the quaternary structure of α3 NC1 domain of collagen IV. Indeed, anti-GBM disease may be considered an autoimmune "conformeropathy" that involves the perturbation of quaternary structure of the α 345NC1 hexamer. MHC-restricted T cells initiate the autoantibody response because humans are not tolerant to the epitopes created by this quaternary structure. The epitopes are normally sequestered in the collagen IV hexamer and can be exposed by infection, smoking, oxidants, or solvents. Goodpasture's syndrome appears in two age groups: in young men in their late twenties and in men and women in their sixties and seventies. Younger patients are more likely to present with the full Goodpasture's syndrome, with hemoptysis, a sudden fall in hemoglobin, fever, dyspnea, and hematuria, and older patients are more likely to present with isolated glomerulonephritis. Hemoptysis is largely confined to smokers, and those who present with lung hemorrhage as a group do better than older populations who have prolonged, asymptomatic renal injury;

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presentation with oliguria is often associated with a particularly bad outcome. The performance of an urgent kidney biopsy is important in suspected cases of Goodpasture's syndrome to confirm the diagnosis and assess prognosis. Renal biopsies typically show *focal or segmental necrosis* that later, with aggressive destruction of the capillaries by cellular proliferation, leads to crescent formation in Bowman's space **(see Fig. A4-14)**. As these lesions progress, there is concomitant interstitial nephritis with fibrosis and tubular atrophy.

The presence of anti-GBM antibodies and complement is recognized on biopsy by linear immunofluorescent staining for IgG (rarely IgA). In testing serum for anti-GBM antibodies, it is particularly important that the α3 NC1 domain of collagen IV alone be used as the target. This is because nonnephritic antibodies against the α1 NC1 domain are seen in paraneoplastic syndromes and cannot be discerned from assays that use whole basement membrane fragments as the binding target. Between 10 and 15% of sera from patients with Goodpasture's syndrome also contain ANCA antibodies against myeloperoxidase. Prognosis at presentation is worse if there are >50% crescents on renal biopsy with advanced fibrosis, if serum creatinine is >5–6 mg/dL, if oliguria is present, or if there is a need for acute dialysis. Patients who present with hemoptysis should be treated for their lung hemorrhage, as it responds to plasmapheresis. Treated patients with less severe disease typically respond to 8–10 treatments of plasmapheresis accompanied by oral prednisone and cyclophosphamide. Maintenance therapy with low-dose immunosuppressants should be considered until antibody titers are negative. There are scarce data alternatively using rituximab, azathioprine, or mycophenolate mofetil. Kidney transplantation should wait for 6 months and until serum antibodies are undetectable.

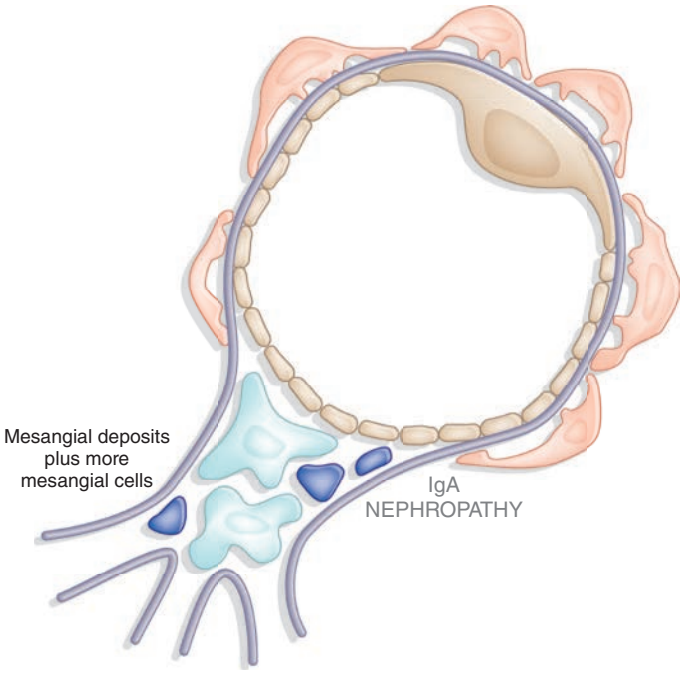
IgA NEPHROPATHY

Berger first described the glomerulonephritis now termed *IgA nephropathy*. It is classically characterized by episodic hematuria associated with the deposition of IgA in the mesangium. IgA nephropathy is one of the most common forms of glomerulonephritis worldwide. There is a male preponderance, a peak incidence in the second and third decades of life, and rare familial clustering. There are geographic differences in the prevalence of IgA nephropathy, with 30% prevalence along the Asian and Pacific Rim and 20% in southern Europe, compared to a much lower prevalence in northern Europe and North America. This may reflect variation in detection or a true variation among racial and ethnic groups.

IgA nephropathy is predominantly a sporadic disease, but susceptibility to it has been shown uncommonly to have a genetic component depending on geography and the existence of "founder effects." Familial forms of IgA nephropathy are more common in northern Italy and eastern Kentucky. No single causal gene has been identified. Clinical and laboratory evidence suggests close similarities between Henoch-Schönlein purpura and IgA nephropathy. Henoch-Schönlein purpura is distinguished clinically from IgA nephropathy by prominent systemic symptoms, a younger age (<20 years old), preceding infection, and abdominal complaints. Deposits of IgA are also found in the glomerular mesangium in a variety of systemic diseases, including chronic liver disease, Crohn's disease, celiac disease, chronic bronchiectasis, idiopathic interstitial pneumonia, dermatitis herpetiformis, mycosis fungoides, ankylosing spondylitis, HIV infection, and Sjögren's syndrome. IgA deposition in these entities is not usually associated with clinically significant renal disease. IgG A-dominant *Staphylococcus*-associated postinfectious glomerulonephritis is associated with clinically significant renal disease.

The pathognomonic finding on kidney biopsy is dominant or codominant mesangial IgA deposits, either alone or with IgG, IgM, or C_3. **(See Glomerular Schematic 2.)** IgA deposits are typically J-chain containing polymeric IgA. Abnormalities have been described in IgA production by plasma cells, in IgA clearance by the liver, and in mesangial IgA clearance and receptors for IgA. Currently, however, abnormalities in the *O*-glycosylation of the hinge region of primarily polymeric IgA1 seem to best account for the pathogenesis of sporadic IgA nephropathy. Synthesis of poorly galactosylated IgA1 results in exposure of *N*-acetyl-galactosamine in truncated IgA1 hinge regions,

Glomerular schematic 2



which is recognized by IgG or IgA1 antibodies leading to formation of immune complexes in the circulation or in situ after glomerular deposition of galactose-deficient IgA1. A second hit, such as a viral or other antigen exposure, or hereditary defects in alternative complement pathway proteins may affect the manifestation of disease. Despite the presence of elevated serum IgA levels in 20–50% of patients and IgA deposition in skin biopsies in 15–55% of patients, a renal biopsy is necessary to confirm the diagnosis. Although the immunofluorescent pattern of IgA on renal biopsy defines IgA nephropathy in the proper clinical context, a variety of histologic lesions may be seen on light microscopy **(see Fig. A4-8)**, including DPGN; *segmental sclerosis*; and, rarely, *segmental necrosis with cellular crescent formation*, which typically presents as RPGN.

The two most common presentations of IgA nephropathy are recurrent episodes of macroscopic hematuria during or immediately following an upper respiratory infection often accompanied by proteinuria and persistent asymptomatic microscopic hematuria. Nephrotic syndrome is uncommon. Proteinuria can also first appear late in the course of the disease. Rarely, patients present with acute renal failure and a rapidly progressive clinical picture. IgA nephropathy is a benign disease for the majority of patients, and 5–30% of patients may go into a complete remission, with others having hematuria but well-preserved renal function. In the minority of patients who have progressive disease, progression is slow, with renal failure seen in only 25–30% of patients with IgA nephropathy over 20–25 years. This risk varies considerably among populations. Cumulatively, risk factors for the loss of renal function identified thus far account for <50% of the variation in observed outcome but include the presence of hypertension or proteinuria, the absence of episodes of macroscopic hematuria, male sex, and older age of onset. Mesangial hypercellularity (M), endocapillary hypercellularity (E), segmental glomerulosclerosis (S), tubular interstitial fibrosis (T), and crescents (C) have predictive value as established by the Oxford Classification and the MEST-C score. Several analyses in large populations of patients found persistent proteinuria for 6 months or longer to have the greatest predictive power for adverse renal outcomes.

There is no agreement on optimal treatment. Both large studies that include patients with multiple glomerular diseases and small studies of patients with IgA nephropathy support the use of angiotensin-converting enzyme (ACE) inhibitors in patients with proteinuria or declining renal function. In patients with persistent proteinuria after

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ACE inhibitor therapy, steroid treatment or other immunosuppressives have demonstrated conflicting results. Tonsillectomy and fish oil have also been suggested in small studies to benefit select patients. When presenting as RPGN, patients typically receive steroids, cytotoxic agents, and plasmapheresis.

ANCA SMALL-VESSEL VASCULITIS

A group of patients with small-vessel vasculitis (arterioles, capillaries, and venules; rarely small arteries) and glomerulonephritis have serum ANCA; the antibodies are of two types, anti-proteinase 3 (PR3) or anti-myeloperoxidase (MPO) **(Chap. 363)**. ANCA are produced with the help of T cells and activate leukocytes and monocytes, which together damage the walls of small vessels. Endothelial injury also attracts more leukocytes and extends the inflammation. Granulomatosis with polyangiitis, microscopic polyangiitis, Churg-Strauss syndrome, and renal-limited vasculitis belong to this group because they are ANCA-positive and have a *pauci-immune glomerulonephritis* with few immune complexes in small vessels and glomerular capillaries. Patients with any of these diseases can have any combination of the above serum antibodies, but anti-PR3 antibodies are more common in granulomatosis with polyangiitis, and anti-MPO antibodies are more common in microscopic polyangiitis or Churg-Strauss. Although each of these diseases has some unique clinical features, most features do not predict relapse or progression, and as a group, they are generally treated in the same way. Once diagnosed, ANCA monitoring has limited value, but targeted determination of ANCA levels may be useful if a relapse is clinically suspected. Since mortality is high without treatment, virtually all patients receive urgent treatment. Induction therapy usually includes glucocorticoids and either cyclophosphamide or rituximab. Plasmapheresis is recommended in rapidly progressive renal failure or pulmonary hemorrhage. Remission is induced in 85–90% of patients, but relapse is common. Steroids are tapered soon after acute inflammation subsides. Maintenance therapy includes low-dose steroids and cyclophosphamide or less toxic agents such as azathioprine, methotrexate, or rituximab for up to a year to minimize the risk of relapse.

Granulomatosis with Polyangiitis Patients with this disease classically present with fever, purulent rhinorrhea, nasal ulcers, sinus pain, polyarthralgias/arthritis, cough, hemoptysis, shortness of breath, hematuria, and subnephrotic proteinuria; occasionally, there may be cutaneous purpura and mononeuritis multiplex. Patients may present without renal involvement, although most of these patients develop renal injury later. Chest x-ray often reveals nodules and persistent infiltrates, sometimes with cavities. Biopsy of involved tissue will show a small-vessel vasculitis and adjacent noncaseating granulomas. Renal biopsies during active disease demonstrate *segmental necrotizing glomerulonephritis* without immune deposits and have been classified as focal, mixed, crescentic, or sclerotic **(see Fig. A4-13)**. The disease is more common in patients exposed to silica dust and those with α_1-antitrypsin deficiency, which is an inhibitor of PR3. Relapse after achieving remission is common and is more common in patients with granulomatosis with polyangiitis than the other ANCA-associated vasculitis, necessitating diligent follow-up care. Although associated with an unacceptable high mortality rate without treatment, the greatest threat to patients, especially elderly patients in the first year of therapy, is from adverse events, which are often secondary to treatment, rather than active vasculitis. Patients should also be monitored long term for malignancy after immunosuppressive therapy.

Microscopic Polyangiitis Clinically, these patients look somewhat similar to those with granulomatosis with polyangiitis, except they rarely have significant lung disease or destructive sinusitis. The distinction is made on biopsy, where the vasculitis in microscopic polyangiitis is without granulomas. Some patients will also have injury limited to the capillaries and venules.

Churg-Strauss Syndrome When small-vessel vasculitis is associated with peripheral eosinophilia, cutaneous purpura, mononeuritis, asthma, and allergic rhinitis, a diagnosis of Churg-Strauss syndrome (eosinophilic granulomatosis with polyangiitis [EGPA]) is considered. Hypergammaglobulinemia, elevated levels of serum IgE, or the presence of rheumatoid factor sometimes accompanies the allergic state. Lung inflammation, including fleeting cough and pulmonary infiltrates, often precedes the systemic manifestations of disease by years; lung manifestations are rarely absent. A third of patients may have exudative pleural effusions associated with eosinophils. Small-vessel vasculitis and *focal segmental necrotizing glomerulonephritis* without immune deposits can be seen on renal biopsy, usually absent eosinophils or granulomas. The cause of Churg-Strauss syndrome is autoimmune, but the inciting factors are unknown.

C_3 GLOMERULOPATHIES

C_3 glomerulopathy is a recent disease classification that is defined by the glomerular accumulation of C_3 with little or no immunoglobulin and encompasses dense deposit disease (DDD), formerly MPGN type II (see below), and C_3 glomerulonephritis (C_3GN) **(Table 314-4)**. DDD is defined morphologically by dense deposits forming ribbons in the GBM. In the absence of this specific morphology, the entity is categorized as C_3GN. Both are associated with the presence of a complement mutation believed to cause the renal pathology, including mutations in the complement factor H regulatory (CFHR) protein genes. DDD is primarily a disease of children and young adults, whereas the other C_3 glomerulopathies are reported to present in an older age group (mean age 30). By definition, kidneys with C_3 glomerulopathy show sole or dominant staining for C_3 but can have variable light microscopy, with mesangial proliferative or membranoproliferative patterns seen most commonly. Morphologically, many cases are not distinguishable from recovering postinfectious glomerulonephritis. Patients with DDD present with proteinuria, which may be nephrotic range, and/or hematuria, which may be macroscopic or microscopic. Partial lipodystrophy and Drusen bodies in the retina may also be present. Prognosis is poor, with 50% of patients progressing to ESRD. C_3GN patients are clinically less well defined, but approximately two-thirds have hematuria and one-third have proteinuria. C_3 levels are low with normal C_4, and C_3 nephritic factor is present in most patients with DDD and less commonly in C_3GN. Abnormalities in factor H, soluble C5b-9, paraprotein detection, and specific CFHR genetic mutations should be assessed as well. Screening family members may be indicated. The optimal therapies remain undefined but include inhibition of the renin-angiotensin

TABLE 314-4 Membranoproliferative Glomerulonephritis: Immunoglobulin-Mediated

Type I Disease—Most Common
Idiopathic
Infection: Subacute bacterial endocarditis, hepatitis B and C, fungal and parasitic infections
Autoimmune diseases: Systemic lupus erythematosus, cryoglobulinemia, Sjögren's
Monoclonal gammopathies: Monoclonal gammopathy of undetermined significance, myeloma, monoclonal gammopathy of renal significance
Cancer: Lung, breast, and ovary (germinal)
Type II Disease
Idiopathic
Dense deposit disease (immunoglobulin-mediated)
Type III Disease
Idiopathic
C_3 Glomerulopathy: C_3 Dominant, Non-Immunoglobulin-Mediated
Dense Deposit Disease (C_3 dominant)
Idiopathic
Specific genetic mutations and/or autoantibodies to alternate complement pathway factors or regulatory factors of alternate complement pathway
C_3 Glomerulonephritis
Specific genetic mutations and/or autoantibodies to alternate complement pathway factors or regulatory factors of alternate complement pathway

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system, lipid lowering, steroids, and other immunosuppressants. Evidence suggests a benefit of therapy with eculizumab, a monoclonal antibody directed at C_5, which is activated by C_3.

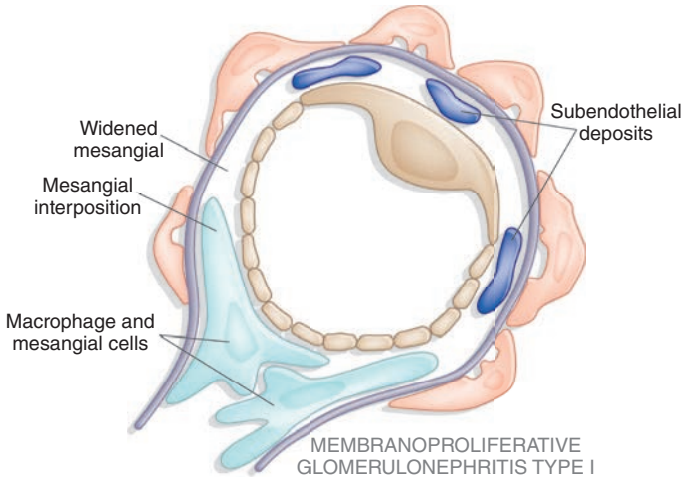
MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS

MPGN is characterized by thickening of the GBM with mesangioproliferative changes often leading to a lobular appearance of the glomerular tuft; 70% of patients have hypocomplementemia. MPGN is rare in African Americans, and idiopathic disease usually presents in childhood or young adulthood. MPGN has been subdivided based on histology into type I, type II, and type III disease. *Type I MPGN* is immune complex–mediated and commonly associated with persistent hepatitis B and C, fungal and parasitic infections, SBE, autoimmune diseases such as lupus or cryoglobulinemia, or monoclonal gammopathies, including monoclonal gammopathy of renal significance (MGRS), where the only clinically apparent manifestations are in the kidney (Table 314-4). *Types II and III MPGN* can be idiopathic and immunoglobulin-mediated disease (driven by the classical complement pathway), but the vast majority of cases formerly defined as MPGN type II or III are non-immunoglobulin-mediated and driven by the alternate complement pathway.

Type I MPGN, the most proliferative of the three types, shows mesangial proliferation with lobular segmentation on renal biopsy and mesangial interposition between the capillary basement membrane and endothelial cells, producing a double contour sometimes called *tram-tracking* **(see Fig. A4-9). (See Glomerular Schematic 3.)** Subendothelial deposits with low serum levels of C_3 are typical, although 50% of patients have normal levels of C_3 and occasional intramesangial deposits. Low serum C_3 and a dense thickening of the GBM containing ribbons of dense deposits and C_3 characterize type II MPGN, *dense deposit disease* **(see Fig. A4-10)**. Classically, the glomerular tuft has a lobular appearance; intramesangial deposits are rarely present, and subendothelial deposits are generally absent. Proliferation in type III MPGN is less common than the other two types and is often focal; mesangial interposition is rare, and subepithelial deposits as well as subendothelial deposits can occur along widened segments of the GBM that appear laminated and disrupted.

Classic type I MPGN is secondary to glomerular deposition of circulating immune complexes or their in situ formation. Patients with MPGN present with proteinuria, hematuria, and pyuria (30%); systemic symptoms of fatigue and malaise that are most common in children with type I disease; or an acute nephritic picture with RPGN and a speedy deterioration in renal function in up to 25% of patients. Low serum C_3 levels are common. Fifty percent of patients with MPGN develop ESRD 10 years after diagnosis, and 90% have renal insufficiency after 20 years. Nephrotic syndrome, hypertension, and renal insufficiency all predict poor outcome. In the presence of proteinuria, treatment with inhibitors of the renin-angiotensin system is prudent. Evidence supports the efficacy of treatment of *primary MPGN* with steroids, particularly in children. There are reports of efficacy with other immunosuppressive drugs. If defects in the complement pathway are found, treatment with eculizumab is of benefit. In *secondary MPGN,* treating the associated infection, autoimmune disease, or neoplasms is of demonstrated benefit. Patients with primary MPGN are well known to be at risk for not only a histologic recurrence in the transplanted kidney but also a clinically significant recurrence with loss of graft function.

Glomerular schematic 3



MESANGIOPROLIFERATIVE GLOMERULONEPHRITIS

Mesangioproliferative glomerulonephritis is characterized by expansion of the mesangium, sometimes associated with mesangial hypercellularity; thin, single contoured capillary walls; and mesangial immune deposits. Mesangioproliferative pathology may be seen in IgA nephropathy, *Plasmodium falciparum* malaria, resolving postinfectious glomerulonephritis, and class II nephritis from lupus, all of which can have a similar histologic appearance. With these secondary entities excluded, the diagnosis of *primary mesangioproliferative glomerulonephritis* is made in <15% of renal biopsies.

NEPHROTIC SYNDROME

Nephrotic syndrome classically presents with heavy proteinuria, minimal hematuria, hypoalbuminemia, hypercholesterolemia, edema, and hypertension. If left undiagnosed or untreated, some of these syndromes will progressively damage enough glomeruli to cause a fall in GFR, producing renal failure. Multiple studies have noted that the higher the 24-h urine protein excretion, the more rapid is the decline in GFR.

Therapies for various causes of nephrotic syndrome are noted under individual disease headings below. In general, all patients with hypercholesterolemia secondary to nephrotic syndrome should be treated with lipid-lowering agents because they are at increased risk for cardiovascular disease. Edema secondary to salt and water retention can be controlled with the use of diuretics, avoiding intravascular volume depletion. Venous complications secondary to the hypercoagulable state associated with nephrotic syndrome can be treated with anticoagulants. The losses of various serum binding proteins, such as thyroid-binding globulin, lead to alterations in functional tests. Lastly, proteinuria itself is hypothesized to be nephrotoxic, and treatment of proteinuria with inhibitors of the renin-angiotensin system can lower urinary protein excretion.

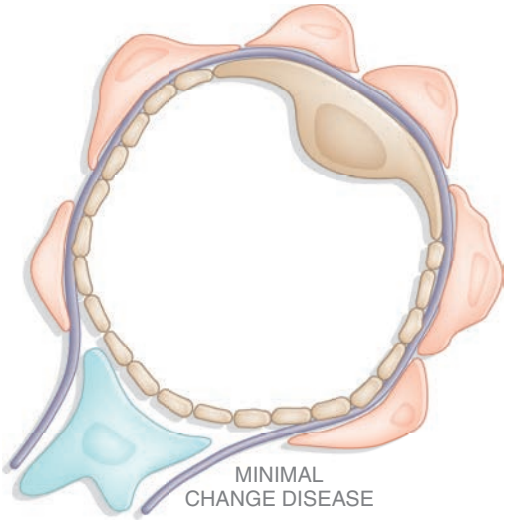
MINIMAL CHANGE DISEASE

MCD, sometimes known as *nil lesion,* causes 70–90% of nephrotic syndrome in childhood but only 10–15% of nephrotic syndrome in adults. MCD usually presents as a primary renal disease but can be associated with several conditions, including Hodgkin's disease, allergies, use of nonsteroidal anti-inflammatory agents or lithium, infections, and other glomerular diseases. MCD on renal biopsy shows no glomerular lesion by light microscopy and is negative for deposits by immunofluorescent microscopy or occasionally shows small amounts of IgM in the mesangium **(see Fig. A4-1). (See Glomerular Schematic 4.)** Electron microscopy, however, consistently demonstrates an effacement of the foot processes supporting the epithelial podocytes with weakening of slit-pore membranes. The pathogenesis of this lesion is unclear. Most agree there is a disturbance related to T-cell responses, or expression of CD80 or CD40/40L may alter capillary charge and podocyte integrity; interestingly, the use of checkpoint inhibitors as chemotherapy is associated with MCD. There also is some circumstantial evidence for the presence of preceding allergies, altered cell-mediated immunity during viral infections, and a high frequency of remissions with steroids.

MCD presents clinically with the abrupt onset of edema and nephrotic syndrome accompanied by acellular urinary sediment. Average urine protein excretion reported in 24 h is 10 g with severe hypoalbuminemia. Less common clinical features include hypertension (30% in children, 50% in adults), microscopic hematuria (20% in children,

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Glomerular schematic 4



33% in adults), atopy or allergic symptoms (40% in children, 30% in adults), and decreased renal function (25–40%), which often returns to normal after remission of the nephrotic syndrome. The appearance of acute renal failure in adults is often seen more commonly in patients with low serum albumin and intrarenal edema (nephrosarca) that is responsive to diuretics. This presentation must be distinguished from acute renal failure secondary to hypovolemia. Acute tubular necrosis and interstitial inflammation are also reported. In children, the abnormal urine principally contains albumin with minimal amounts of higher-molecular-weight proteins and is sometimes called *selective proteinuria*. Although up to 30% of children have a spontaneous remission, most children today are treated with steroids; only children who are nonresponders are biopsied. Primary responders are patients who have a complete remission (<0.2 mg/24 h of proteinuria), often abruptly after a single course of prednisone; steroid-dependent patients relapse as their steroid dose is tapered. Frequent relapsers have two or more relapses in the 6 months following taper, and steroid-resistant patients fail to respond to steroid therapy. Adults are not considered steroid-resistant until after 4 months of therapy. Ninety to 95% of children will develop a complete remission after 8 weeks of steroid therapy, and 80–85% of adults will achieve complete remission, but only after a longer course of 20–24 weeks. Patients with steroid resistance may have FSGS on repeat biopsy. If the first renal biopsy does not have a sample of deeper corticomedullary glomeruli, then the correct diagnosis of FSGS may be missed.

Relapses occur in 70–75% of children after the first remission, and early relapse predicts multiple subsequent relapses, as do high levels of basal proteinuria. The frequency of relapses decreases after puberty. There is an increased risk of relapse following the rapid tapering of steroids in all groups. Relapses are less common in adults but are more resistant to subsequent therapy. Prednisone is first-line therapy, either given daily or on alternate days. Other immunosuppressive drugs, such as cyclophosphamide, chlorambucil, and mycophenolate mofetil, are saved for frequent relapsers, steroid-dependent patients, or steroid-resistant patients. Cyclosporine can induce remission, but relapse is also common when cyclosporine is withdrawn. The long-term prognosis in adults is less favorable when acute renal failure or steroid resistance occurs.

FOCAL SEGMENTAL GLOMERULOSCLEROSIS

FSGS refers to a pattern of renal injury characterized by segmental glomerular scars that involve some but not all glomeruli (focal); the clinical findings of FSGS largely manifest as proteinuria. When the secondary and genetic causes of FSGS are eliminated **(Table 314-5)**, the remaining patients are considered to have primary FSGS. The incidence of this disease is increasing, and it now represents up to one-third of cases of nephrotic syndrome in adults and one-half of cases of nephrotic syndrome in African Americans. The pathogenesis of FSGS has multiple possible mechanisms including a circulating permeability factor, an adaptive response to glomerular hypertrophy or hyperfiltration, and podocyte abnormalities associated with direct toxic injury or genetic mutations. Risk polymorphisms at the *APOL1* locus expressed in podocytes substantially explain the increased burden of FSGS among African Americans.

TABLE 314-5 Focal Segmental Glomerulosclerosis

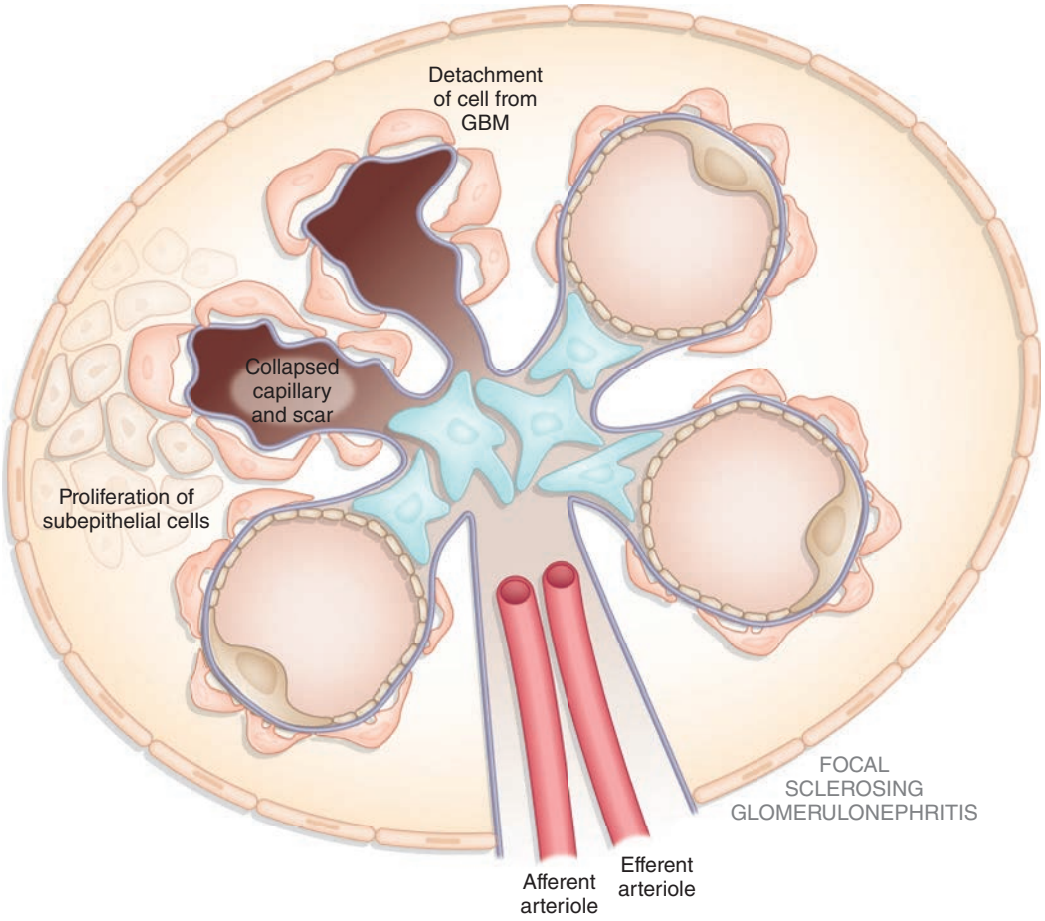
Primary focal segmental glomerulosclerosis
Secondary focal segmental glomerulosclerosis
Adaptive response to hyperfiltration/reduced renal mass, obesity
Viruses: HIV/hepatitis B/parvovirus
Hypertensive nephropathy
Reflux nephropathy
Cholesterol emboli
Drugs: Heroin/analgesics/bisphosphonates/ecstasy
Oligomeganephronia
Sickle cell disease
Radiation nephritis
Familial podocytopathies
NPHS1 mutation/nephrin
NPHS2 mutation/podocin
PLCE1 mutation/phospholipase Cε1
INF2 mutation/inverted formin 2
WT1 mutation/Wilms tumor
TRPC6 mutation/cation channel
ACTN4 mutation/actinin
α-Galactosidase A deficiency/Fabry's disease
N-Acetylneuraminic acid hydrolase deficiency/nephrosialidosis

The pathologic changes of FSGS are most prominent in glomeruli located at the corticomedullary junction **(see Fig. A4-2)**, so if the renal biopsy specimen is from superficial tissue, the lesions can be missed, which sometimes leads to a misdiagnosis of MCD. In addition to focal and segmental scarring, other variants have been described, including cellular lesions with *endocapillary hypercellularity* and heavy proteinuria; *collapsing glomerulopathy* **(see Fig. A4-3)** with segmental or global glomerular collapse and a rapid decline in renal function; a hilar stalk lesion **(see Fig. A4-4)**; or the *glomerular tip lesion* **(see Fig. A4-5)**, which may have a better prognosis. **(See Glomerular Schematic 5.)**

FSGS can present with hematuria, hypertension, any level of proteinuria, and renal insufficiency. Nephrotic-range proteinuria, African-American race, and renal insufficiency are associated with a poor outcome, with 50% of patients reaching renal failure in 6–8 years. FSGS rarely remits spontaneously, but treatment-induced remission of proteinuria significantly improves prognosis. Treatment of patients with *FSGS* should include inhibitors of the renin-angiotensin system. Patients with primary FSGS with nephrotic-range proteinuria can be treated with steroids but respond far less often and after a longer course of therapy than patients with MCD. Proteinuria remits in only 20–45% of patients receiving a course of steroids over 6–12 months. Limited evidence suggests the use of cyclosporine in steroid-responsive patients helps ensure remissions. Relapse frequently occurs after cessation of cyclosporine therapy, and cyclosporine itself can lead to a deterioration of renal function due to its nephrotoxic effects. A role for other agents that suppress the immune system such as rituximab or mycophenolate mofetil has not been established. Immunosuppressive therapy is not indicated in secondary or genetic FSGS. FSGS recurs in 30% of renal transplants, more commonly in primary FSGS, less commonly in secondary FSGS, and rarely in genetic FSGS. In recurrent posttransplant FSGS, many patients will achieve a full or partial remission with plasmapheresis. The treatment of *secondary FSGS* typically involves treating the underlying cause and controlling proteinuria. There is no role for steroids or other immunosuppressive agents in secondary FSGS.

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Glomerular schematic 5



MEMBRANOUS GLOMERULONEPHRITIS

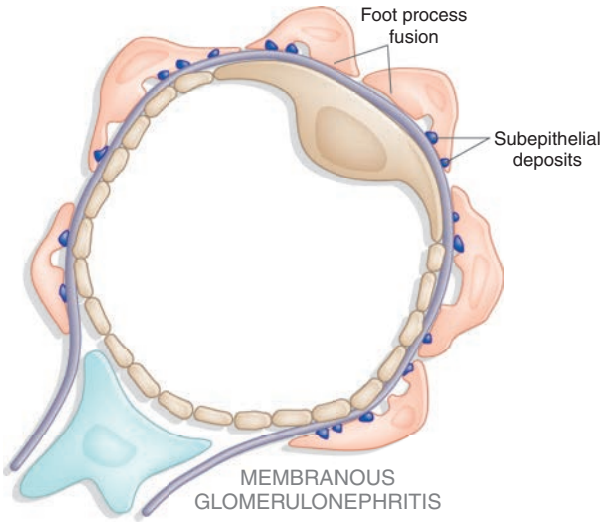
MGN, or *membranous nephropathy* as it is sometimes called, accounts for ~25% of cases of nephrotic syndrome in adults, with a peak incidence between the ages of 30 and 50 years and a male-to-female ratio of 2:1. IMN is rare in childhood and the most common cause of nephrotic syndrome in the elderly. In 20–30% of cases, MGN is secondary and is associated with a malignancy (solid tumors of the breast, lung, colon), infection (hepatitis B, syphilis, malaria, schistosomiasis), rheumatologic disorders such as lupus or rheumatoid arthritis, IgG4 diseases, or drug exposure **(Table 314-6)**.

Uniform thickening of the basement membrane along the peripheral capillary loops is seen by light microscopy on renal biopsy **(see Fig. A4-7)**; this thickening needs to be distinguished from that seen in diabetes and amyloidosis. **(See Glomerular Schematic 6.)** Immunofluorescence demonstrates diffuse granular deposits of IgG and C_3, and electron microscopy typically reveals electron-dense subepithelial deposits. While different stages (I–V) of progressive membranous lesions have been described, some published analyses indicate the degree of tubular atrophy or interstitial fibrosis is more predictive of progression than is the stage of glomerular disease. The presence of subendothelial deposits or the presence of tubuloreticular inclusions strongly points to a diagnosis of membranous lupus nephritis, which may precede the extrarenal manifestations of lupus. In 70% of cases of IMN, autoantibodies against the M-type phospholipase A_2

TABLE 314-6 Membranous Glomerulonephritis

Primary/idiopathic membranous glomerulonephritis
Secondary membranous glomerulonephritis
Infection: Hepatitis B and C, syphilis, malaria, schistosomiasis, leprosy, filariasis
Cancer: Breast, colon, lung, stomach, kidney, esophagus, neuroblastoma
Drugs: Gold, mercury, penicillamine, nonsteroidal anti-inflammatory agents, probenecid, antitumor necrosis factor agents
Autoimmune diseases: Systemic lupus erythematosus, rheumatoid arthritis, primary biliary cirrhosis, dermatitis herpetiformis, bullous pemphigoid, myasthenia gravis, Sjögren's syndrome, Hashimoto's thyroiditis
Other systemic diseases: Fanconi's syndrome, sickle cell anemia, diabetes, Crohn's disease, sarcoidosis, Guillain-Barré syndrome, Weber-Christian disease, angiofollicular lymph node hyperplasia, IgG4 disease

Glomerular schematic 6



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receptor circulate and bind to a conformational epitope present in the PLA2R on human podocytes, producing characteristic in situ deposits. Three to 10% of IMN patients alternatively have autoantibodies to thrombospondin type-1 domain containing 7A (THSD7A). Both antigens co-localize within glomerular subepithelial deposits with IgG4 (PLA2R). Other renal diseases do not involve these autoantibodies. In most cases of secondary membranous nephropathy, these autoantibodies are absent, with rare reports of autoantibodies to PLA2R in membranous glomerulopathy associated with hepatitis B, cancer, and sarcoidosis. Circulating deposits and glomerular deposits of these autoantibodies have correlated with the likelihood of a spontaneous remission, severity of IMN, and the response to therapy. Eighty percent of patients with MGN present with nephrotic syndrome and nonselective proteinuria. Microscopic hematuria is seen but less commonly than in IgA nephropathy or FSGS. Spontaneous remissions occur in 20–33% of patients and often occur late in the course, which makes treatment decisions difficult. One-third of patients continue to have relapsing nephrotic syndrome but maintain normal renal function, and approximately another third of patients develop renal failure or die from the complications of nephrotic syndrome. Male gender, older age, hypertension, and the persistence of nephrotic-range proteinuria are associated with worse prognosis. Although thrombotic complications are a feature of all nephrotic syndromes, MGN has the highest reported incidences of renal vein thrombosis, pulmonary embolism, and deep-vein thrombosis. Prophylactic anticoagulation is controversial but has been recommended for patients with hypoalbuminemia.

In addition to the treatment of edema, dyslipidemia, and hypertension, inhibition of the renin-angiotensin system is recommended. Therapy with immunosuppressive drugs is also recommended for patients with primary MGN and persistent proteinuria (>3.0 g/24 h). The choice of immunosuppressive drugs for therapy is controversial, but current recommendations are to treat with steroids and cyclophosphamide, chlorambucil, mycophenolate mofetil, or cyclosporine or rituximab, an anti-CD20 antibody directed at B cells. Attaining remission is associated with a good long-term prognosis.

DIABETIC NEPHROPATHY

Diabetic nephropathy is the single most common cause of chronic renal failure in the United States and worldwide. The dramatic increase in the number of patients with diabetic nephropathy reflects the epidemic increase in obesity and type 2 diabetes mellitus. Approximately 40% of patients with diabetes develop nephropathy, but due to the higher prevalence of type 2 diabetes (90%) compared to type 1 (10%), the majority of patients with diabetic nephropathy have type 2 disease. Renal lesions are more common in African-American, Native-American, Polynesian, and Maori populations. Risk factors for the development of diabetic nephropathy include hyperglycemia, hypertension, dyslipidemia, smoking, a family history of diabetic nephropathy, and gene polymorphisms.

Within 1–2 years after the onset of clinical diabetes, morphologic changes appear in the kidney. Thickening of the GBM is a sensitive indicator for the presence of diabetes but correlates poorly with the presence or absence of nephropathy. The composition of the GBM is altered notably with a loss of heparan sulfate moieties that form the negatively charged filtration barrier resulting in increased filtration of serum proteins into the urine. The expansion of the mesangium due to the accumulation of extracellular matrix correlates with the clinical manifestations of diabetic nephropathy **(see stages in Fig. A4-20)**. This expansion in mesangial matrix is associated with the development of *mesangial sclerosis*. Some patients also develop eosinophilic, PAS^+ nodules called *nodular glomerulosclerosis* or *Kimmelstiel-Wilson nodules*. Immunofluorescence microscopy often reveals the nonspecific deposition of IgG (at times in a linear pattern) or complement staining without immune deposits on electron microscopy. Prominent vascular changes are frequently seen with hyaline and hypertensive arteriosclerosis. This is associated with varying degrees of chronic glomerulosclerosis and tubulointerstitial changes. Renal biopsies from patients with types 1 and 2 diabetic nephropathies with albuminuria are largely indistinguishable. Patients with type 2 diabetes without albuminuria are classified as having diabetic kidney disease as opposed to diabetic nephropathy and may have myriad pathologic findings.

Multiple lines of evidence support an important role for changes in glomerular hemodynamics including increases in glomerular capillary pressure and glomerular hyperfiltration in these pathologic changes. Hyperglycemia activates the renin-angiotensin-aldosterone system and also alters insulin-like growth factor, reactive oxygen species, and endothelin 1. Diabetes upregulates the sodium-glucose cotransporters (SGLT1 and SGLT2) in the proximal tubule, resulting in decreased distal delivery of sodium to the macula densa and further glomerular hyperfiltration. Sustained glomerular hypertension increases matrix production and alterations in the GBM with disruption in the filtration barrier. Other factors that alter matrix production include the accumulation of advanced glycosylation end products, circulating factors including growth hormone, connective tissue growth factor, TGF-β, and dyslipidemia.

The natural history of diabetic nephropathy has been historically well characterized in the ~40% of diabetics who develop it as a progression from glomerular hyperfiltration and renal hypertrophy to increasing albuminuria followed by declining GFR and ESRD. However, since the onset of type 1 diabetes is readily identifiable and the onset of type 2 diabetes is not, a patient newly diagnosed with type 2 diabetes may present with *advanced diabetic nephropathy*. Albuminuria and decreased GFR are potent risk factors for cardiovascular disease, with some patients dying before they reach ESRD. Furthermore, contemporary studies reveal that up to 24% of patients with type 1 diabetes and 50% with type 2 diabetes and chronic kidney disease may be normoalbuminuric. It is unknown whether this alteration in the natural history reflects contemporary effective interventions or perhaps other kidney diseases that happen to occur in patients with diabetes. The degree of early glomerular hyperfiltration does correlate with the development of albuminuria and declining GFR. Albuminuria in the range of 30–300 mg/24 h is called *microalbuminuria* (Table 314-1). Microalbuminuria appears 5–10 years after the onset of diabetes. It is currently recommended to test patients with type 1 disease for microalbuminuria 5 years after diagnosis of diabetes and yearly thereafter and, because the time of onset of type 2 diabetes is often unknown, to test type 2 patients at the time of diagnosis of diabetes and yearly thereafter. Microalbuminuria classically progresses over 5–10 years to proteinuria and declining GFR, but in contemporary studies, greater heterogeneity is reported with regression to normoalbuminuria; however, albuminuria remains the single most important predictor of a faster decline in GFR. Regression of albuminuria with a treatment intervention is a good prognostic sign. Proteinuria in diabetic nephropathy can be variable, ranging from 500 mg to 25 g/24 h. More than 90% of patients with type 1 diabetes and nephropathy have diabetic retinopathy, so the absence of retinopathy in type 1 patients with proteinuria should prompt consideration of a diagnosis other than diabetic nephropathy; only 60% of patients with type 2 diabetes with nephropathy have diabetic retinopathy. There is a significant correlation between the presence of retinopathy and the presence of Kimmelstiel-Wilson nodules **(see Fig. A4-20)**. Even with advanced chronic kidney disease, patients with diabetic nephropathy will have enlarged kidneys. Using the above data, and in the absence of other clinical or serologic data suggesting another disease, diabetic nephropathy is usually diagnosed without a renal biopsy. The risk of progression to ESRD is influenced by treatment and other risk factors, and reports vary from a decline of 1.8–14 mL/min per year. Survival on dialysis is worse for patients with diabetes. Renal transplantation results in better survival than dialysis.

Good evidence supports the benefits of blood sugar and blood pressure control, inhibitors of the renin-angiotensin-aldosterone system (RAAS), and inhibitors of SGLT2 in retarding the progression of diabetic nephropathy. In patients with type 1 diabetes, intensive control of blood sugar clearly prevents the development or progression of diabetic nephropathy. The evidence for benefit of intensive blood glucose control in patients with type 2 diabetes is less certain, with current studies reporting conflicting results.

Controlling systemic blood pressure decreases renal and cardiovascular adverse events in this high-risk population. The vast majority of

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patients with diabetic nephropathy require three or more antihypertensive drugs to achieve this goal. Drugs that inhibit the RAAS (ACE inhibitors, angiotensin receptor blockers [ARBs]), independent of their effects on systemic blood pressure, have been shown in large clinical trials to slow the progression of diabetic nephropathy at early (microalbuminuria) and late (proteinuria with reduced glomerular filtration) stages. Evidence suggests increased risk for cardiovascular adverse events without increased efficacy in patients with a combination of two drugs (ACE inhibitors, ARBs, or renin inhibitors) that suppress several components of the RAAS. In patients with type 2 diabetes and kidney disease with albuminuria, the risk of kidney failure and cardiovascular events was lower in those receiving SGLT2 in addition to inhibitors of the RAAS. Ongoing trials are examining the hypotheses that other agents may be of benefit, including endothelin antagonists and aldosterone antagonists.

GLOMERULAR DEPOSITION DISEASES

Plasma cell dyscrasias producing excess light chain immunoglobulin sometimes lead to the formation of glomerular and tubular deposits that cause heavy proteinuria and renal failure; the same is true for the accumulation of serum amyloid A protein fragments seen in several inflammatory diseases. This broad group of proteinuric patients has *glomerular deposition disease*.

Light Chain Deposition Disease The biochemical characteristics of nephrotoxic light chains produced in patients with light chain malignancies confer renal injury; that of either *cast nephropathy* **(see Fig. A4-17)**, which causes renal failure but not heavy proteinuria or amyloidosis, or light chain deposition disease (LCDD) **(see Fig. A4-16)**, which produces proteinuria with renal failure. These latter patients produce kappa light chains that do not have the biochemical features necessary to form amyloid fibrils. Instead, they self-aggregate and form granular deposits along the glomerular capillary and mesangium or, more prominently, in the tubular basement membrane and Bowman's capsule. Light chain deposits are not fibrillar and do not stain with Congo red, but they are easily detected with anti–light chain antibody. A combination of the light chain rearrangement, self-aggregating properties at neutral pH, and abnormal metabolism probably contributes to the deposition.

Monoclonal Plasma Cell Disorders Multiple myeloma, Waldenström's macroglobulinemia, or lymphoma may be present, as well as heart, liver, and pulmonary involvement. The monoclonal protein may be found with serum electrophoresis or with serum free light chain analysis. Nephrotic syndrome may develop, and ~70% of patients progress to dialysis. Treatment for LCDD is treatment of the primary disease and, if possible, autologous stem cell transplantation. Rarely, truncated heavy chains similarly cause nonamyloid deposits.

Renal Amyloidosis Most *renal amyloidosis* is either the result of primary fibrillar deposits of immunoglobulin light chains known as amyloid L (AL) or secondary to fibrillar deposits of serum amyloid A (AA) protein fragments **(Chap. 112)**. Even though both occur for different reasons, their clinicopathophysiology is quite similar. Amyloid infiltrates the liver, heart, peripheral nerves, carpal tunnel, upper pharynx, and kidney, producing restrictive cardiomyopathy, hepatomegaly, macroglossia, and heavy proteinuria sometimes associated with renal vein thrombosis. In contrast to LCDD, amyloid renal deposits are fibrillar, stain with Congo red, and contain predominantly the variable region of lambda chains **(see Fig. A4-15)**. In systemic AL amyloidosis, also called *primary amyloidosis,* light chains produced in excess by clonal plasma cell dyscrasias are made into fragments by macrophages so they can self-aggregate at acid pH. Approximately 10% of patients have overt myeloma as defined by CRAB (hypercalcemia, renal insufficiency, anemia, or lytic bone lesions). Nephrotic syndrome is common, and ~20% of patients progress to dialysis. AA amyloidosis is sometimes called *secondary amyloidosis* and also presents as nephrotic syndrome. It is due to deposition of β-pleated sheets of serum amyloid A protein, an acute phase reactant. Patients with AA amyloid have associated inflammatory diseases including rheumatoid arthritis, ankylosing spondylitis, psoriatic arthritis, juvenile inflammatory arthritis, and familial Mediterranean fever. An increasing proportion of patients have unidentified chronic inflammation; this may reflect better treatments for the previously associated diseases or a rise in chronic inflammation due to obesity. Fragments of serum amyloid A protein increase and self-aggregate by attaching to receptors for advanced glycation end products in the extracellular environment; nephrotic syndrome is common, and ~40–60% of patient's progress to dialysis. Serum-free light chain analysis is useful in the early diagnosis and follow-up of disease progression. Biopsy of involved liver or kidney is diagnostic 90% of the time when the pretest probability is high; abdominal fat pad aspirates are positive ~70% of the time, but apparently less so when looking for AA amyloid. Amyloid deposits are distributed along blood vessels and in the mesangial regions of the kidney. The treatment for primary amyloidosis, melphalan, and autologous hematopoietic stem cell transplantation (HCT) can delay the course of disease. Patients who are not candidates for HCT often receive bortezomib-based regimens. Secondary amyloidosis is also relentless unless the primary disease can be controlled. Some new drugs in development that disrupt the formation of fibrils may be available in the future.

Fibrillary and Immunotactoid Glomerulopathies Fibrillary and immunotactoid glomerulopathies are rare (<1.0% of renal biopsies), morphologically defined diseases characterized by glomerular accumulation of nonbranching randomly arranged fibrils that are Congo red negative. Fibrillary glomerulopathy accounts for 85–90% of cases and is identified by the presence of the protein DnaJ heat shock protein family B9 (DNAJB9) in the glomeruli, which is absent in the rarer immunotactoid glomerulopathy. In both, glomerular and mesangial deposits contain oligoclonal or oligotypic immunoglobulins and complement, with 12- to 24-nm fibrils in fibrillary glomerulopathy and >30-nm fibrils organized into microtubules in immunotactoid glomerulopathy. The cause of this "nonamyloid" glomerulopathy is mostly idiopathic; reports of fibrillary glomerulonephritis describe associations with malignancy, autoimmune disease, and monoclonal gammopathy, and immunotactoid glomerulopathy has been associated with lymphoma or plasma cell disorders. Both disorders appear in adults aged 40–80 years old, with moderate to heavy proteinuria (100%), hematuria (70%), renal insufficiency (50%), and a wide variety of histologic lesions, including DPGN, MPGN, MGN, or mesangioproliferative glomerulonephritis. Most patients have disease limited to the kidney. Patients should be screened for associated disorders. Half of patients develop renal failure over a few years. There is no consensus on treatment of this uncommon disorder, although rituximab has been reported to remit proteinuria. These diseases can recur in the renal transplant.

FABRY'S DISEASE

Fabry's disease is an X-linked inborn error of globotriaosylceramide metabolism secondary to deficient lysosomal α-galactosidase A (alpha-Gal A) activity, resulting in excessive intracellular storage of globotriaosylceramide. Affected organs include the vascular endothelium, heart, brain, and kidneys. Classically, Fabry's disease presents in childhood in males with acroparesthesias, angiokeratoma, cornea verticillate, and hypohidrosis. Over time, male patients develop cardiomyopathy, cerebrovascular disease, and renal injury, with an average age of death around 50 years of age. Hemizygotes with hypomorphic mutations sometimes present in the fourth to sixth decade with single-organ involvement. Rarely, dominant-negative alpha-Gal A mutations or female heterozygotes with unfavorable X inactivation present with mild single-organ involvement. Rare females develop severe manifestations including renal failure but do so later in life than males. Renal biopsy reveals enlarged glomerular visceral epithelial cells packed with small clear vacuoles containing globotriaosylceramide; vacuoles may also be found in parietal and tubular epithelia **(see Fig. A4-18)**. These vacuoles of electron-dense materials in parallel arrays (zebra bodies) are easily seen on electron microscopy. Ultimately, renal biopsies reveal FSGS. The nephropathy of Fabry's disease typically presents in the third decade as mild to moderate proteinuria, sometimes with microscopic hematuria or nephrotic syndrome. Urinalysis may reveal oval fat bodies and birefringent glycolipid globules under

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polarized light (Maltese cross). Measurement of alpha-Gal A activity and mutational analysis of the gene is diagnostic, with renal biopsies sometimes helpful. Progression to renal failure occurs by the fourth or fifth decade. Treatment with inhibitors of the renin-angiotensin system is recommended. Treatment with recombinant alpha-Gal A or migalastat, a chaperone that facilitates trafficking of alpha-Gal A, clears microvascular endothelial deposits of globotriaosylceramide from the kidneys, heart, and skin. In patients with advanced organ involvement including chronic kidney disease, progression of disease occurs despite enzyme replacement therapy. Variable responses to enzyme therapy may be due to the occurrence of neutralizing antibodies or differences in uptake of the enzyme. Graft and patient survival following renal transplantation in patients with Fabry's disease are similar to those of other causes of ESRD.

PULMONARY-RENAL SYNDROMES

Several diseases can present with catastrophic hemoptysis and glomerulonephritis associated with varying degrees of renal failure. The usual causes include Goodpasture's syndrome, granulomatosis with polyangiitis, microscopic polyangiitis, Churg-Strauss vasculitis, and, rarely, Henoch-Schönlein purpura or cryoglobulinemia. Each of these diseases can also present without hemoptysis and are discussed in detail earlier in "Acute Nephritic Syndromes." **(See Glomerular Schematic 7.)** Pulmonary bleeding in this setting is life-threatening and often results in airway intubation, and acute renal failure requires dialysis. Diagnosis is difficult initially because biopsies and serologic testing take time. Treatment with plasmapheresis and methylprednisolone is often empirical and temporizing until results of testing are available.

BASEMENT MEMBRANE SYNDROMES

All kidney epithelia, including podocytes, rest on basement membranes assembled into a planar surface through the interweaving of collagen IV with laminins, nidogen, and sulfated proteoglycans. Structural abnormalities in GBM associated with hematuria are characteristic of several familial disorders related to the expression of collagen IV genes. The extended family of collagen IV contains six chains, which are expressed in different tissues at different stages of embryonic development. All epithelial basement membranes early in human development are composed of interconnected triple-helical protomers rich in α1.α1.α2(IV) collagen. Some specialized tissues undergo a developmental switch replacing α1.α1.α2(IV) protomers with an α3.α4.α5(IV) collagen network; this switch occurs in the kidney (glomerular and tubular basement membrane), lung, testis, cochlea, and eye, while an α5.α5.α6(IV) network appears in skin, smooth muscle, and esophagus and along Bowman's capsule in the kidney. This switch probably occurs because the α3.α4.α5(IV) network is more resistant to proteases and ensures the structural longevity of critical tissues. When basement membranes are the target of glomerular disease, they produce moderate proteinuria, some hematuria, and progressive renal failure.

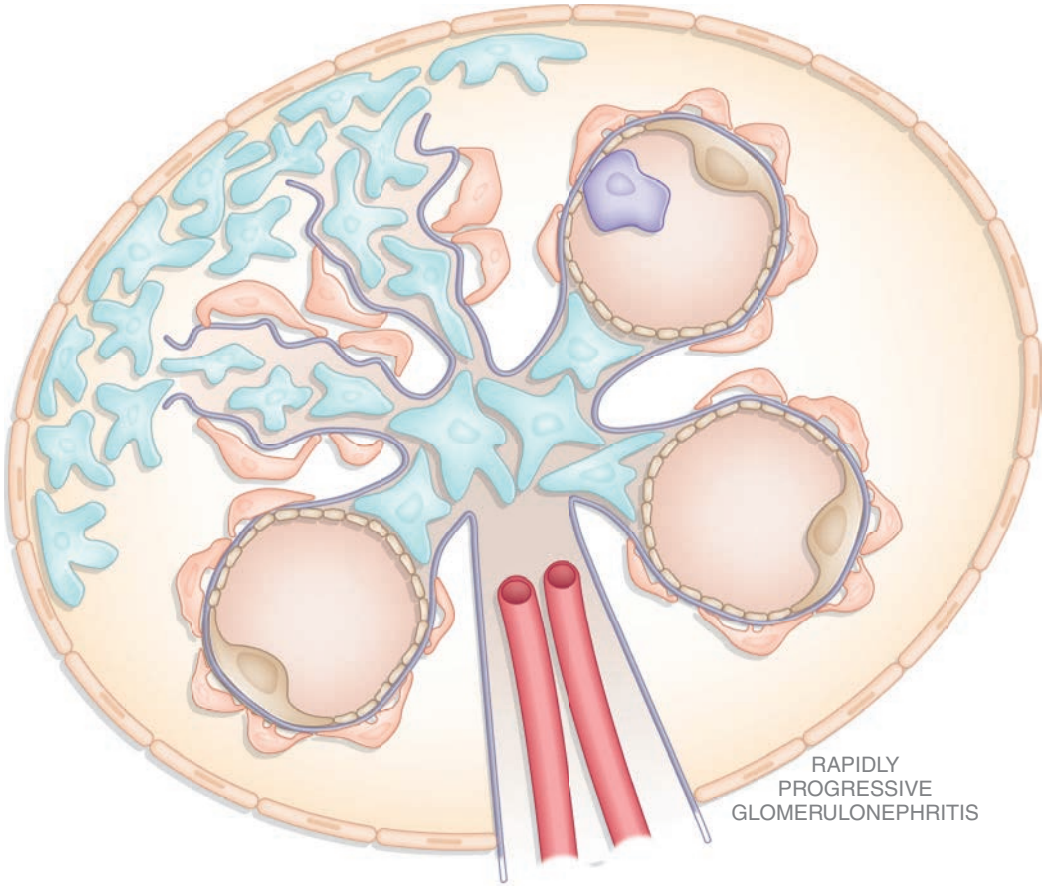
ANTI-GBM DISEASE

Autoimmune disease where antibodies are directed against the α3 NC1 domain of collagen IV produces an *anti-GBM disease* often associated with RPGN and/or a pulmonary-renal syndrome called *Goodpasture's syndrome*. Discussion of this disease is covered earlier in "Acute Nephritic Syndromes."

ALPORT'S SYNDROME

Classically, patients with Alport's syndrome develop hematuria, thinning and splitting of the GBMs, and mild proteinuria (<1–2 g/24 h), which appears late in the course, followed by chronic glomerulosclerosis leading to renal failure in association with sensorineural deafness. Some patients develop lenticonus of the anterior lens capsule, "dot and fleck" retinopathy, and rarely, leiomyomatosis. Approximately 85% of patients with Alport's syndrome have an X-linked inheritance of mutations in the α5(IV) collagen chain on chromosome Xq22–24. Female

Glomerular schematic 7



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carriers have variable penetrance depending on the type of mutation or the degree of mosaicism created by X inactivation. Fifteen percent of patients have autosomal recessive disease of the α3(IV) or α4(IV) chains on chromosome 2q35–37. Rarely, some kindred have an autosomal dominant inheritance of dominant-negative mutations in α3(IV) or α4(IV) chains.

Pedigrees with the X-linked syndrome are quite variable in their rate and frequency of tissue damage leading to organ failure. Seventy percent of patients have the juvenile form with nonsense or missense mutations, reading frame shifts, or large deletions and generally develop renal failure and sensorineural deafness by age 30. Patients with splice variants, exon skipping, or missense mutations of α-helical glycines generally deteriorate after the age of 30 (adult form) with mild or late deafness. Early severe deafness, lenticonus, or proteinuria suggests a poorer prognosis. Usually females from X-linked pedigrees have only microhematuria, but up to 25% of carrier females have been reported to have more severe renal manifestations. Pedigrees with the autosomal recessive form of the disease have severe early disease in both females and males with asymptomatic parents.

Clinical evaluation should include a careful eye examination and hearing tests. However, the absence of extrarenal symptoms does not rule out the diagnosis. Since α5(IV) collagen is expressed in the skin, some X-linked Alport's patients can be diagnosed with a skin biopsy revealing the lack of the α5(IV) collagen chain on immunofluorescent analysis. Patients with mutations in α3(IV) or α4(IV) require a renal biopsy. Genetic testing can be used for the diagnosis of Alport's syndrome and the demonstration of the mode of inheritance. Early in their disease, Alport's patients typically have thin basement membranes on renal biopsy **(see Fig. A4-19)**, which thicken over time into multilamellations surrounding lucent areas that often contain granules of varying density—the so-called split basement membrane. In any Alport's kidney, there are areas of thinning mixed with splitting of the GBM. Tubules drop out, glomeruli scar, and the kidney eventually succumbs to interstitial fibrosis. All affected members of a family with X-linked Alport's syndrome should be identified and followed, including mothers of affected males. Primary treatment is control of systemic hypertension and use of ACE inhibitors to slow renal progression. Although patients who receive renal allografts usually develop anti-GBM antibodies directed toward the collagen epitopes absent in their native kidney, overt Goodpasture's syndrome is rare and graft survival is good.

THIN BASEMENT MEMBRANE DISEASE

Thin basement membrane disease (TBMD), a relatively common disorder characterized by persistent or intermittent, asymptomatic, usually microscopic hematuria and rarely macroscopic hematuria with flank pain, is not typically associated with proteinuria, hypertension, or loss of renal function or extrarenal disease. TBMD is often familial, with pedigrees exhibiting an autosomal dominant pattern. It usually presents in childhood in multiple family members and is also called *benign familial hematuria*. Many cases of TBMD have genetic defects in type IV collagen, but in contrast to Alport's syndrome, the disease behaves as an autosomal dominant disorder that in ~40% of families segregates with the *COL(IV)* α3/*COL(IV)* α4 loci. Mutations in these loci can result in a spectrum of disease, ranging from TBMD to autosomal dominant or recessive Alport's. The GBM shows diffuse thinning compared to normal values for the patient's age in otherwise normal biopsies **(see Fig. A4-19)**. The vast majority of patients have a benign course.

NAIL-PATELLA SYNDROME

Patients with nail-patella syndrome develop iliac horns on the pelvis and dysplasia of the dorsal limbs involving the patella, elbows, and nails, variably associated with neural-sensory hearing impairment, glaucoma, and abnormalities of the GBM and podocytes, leading to hematuria, proteinuria, and FSGS. The syndrome is autosomal dominant, with haploinsufficiency for the LIM homeodomain transcription factor LMX1B; pedigrees are extremely variable in the penetrance for all features of the disease. LMX1B regulates the expression of genes encoding α3 and α4 chains of collagen IV, interstitial type III collagen, podocin, and CD2AP that help form the slit-pore membranes connecting podocytes. Mutations in the LIM domain region of *LMX1B* associate with glomerulopathy in 30–40% of patients and rarely progress to renal failure. Proteinuria or isolated hematuria is discovered throughout life but usually by the third decade. On renal biopsy, pathognomic rarefications containing clusters of collagen fibrils within the lamina densa of the GBM are found, and FSGS may be present. Treatment is nonspecific, but renin-angiotensin system inhibition is recommended. Patients with renal failure do well with transplantation.

GLOMERULAR-VASCULAR SYNDROMES

A variety of diseases result in classic vascular injury to the glomerular capillaries. Most of these processes also damage blood vessels elsewhere in the body. The group of diseases discussed here lead to vasculitis, renal endothelial injury, thrombosis, ischemia, and/or lipid-based occlusions.

ATHEROSCLEROTIC NEPHROPATHY

Aging in the developed world is commonly associated with the occlusion of coronary and systemic blood vessels. The reasons for this include obesity, insulin resistance, smoking, hypertension, and diets rich in lipids that deposit in the arterial and arteriolar circulation, producing local inflammation and fibrosis of small blood vessels. When the renal arterial circulation is involved, the glomerular microcirculation is damaged, leading to *chronic nephrosclerosis*. Patients with GFRs <60 mL/min have more cardiovascular events and hospitalizations than those with higher filtration rates. Several aggressive lipid disorders can accelerate this process, but most of the time, atherosclerotic progression to chronic nephrosclerosis is associated with poorly controlled hypertension. Approximately 10% of glomeruli are normally sclerotic by age 40, rising to 20% by age 60 and 30% by age 80. Serum lipid profiles in humans are greatly affected by *apolipoprotein E* polymorphisms; the E4 allele is accompanied by increases in serum cholesterol and is more closely associated with atherogenic profiles in patients with renal failure. Mutations in E2 alleles, particularly in Japanese patients, produce a specific renal abnormality called *lipoprotein glomerulopathy* associated with glomerular lipoprotein thrombi and capillary dilation.

HYPERTENSIVE NEPHROSCLEROSIS

Systemic hypertension causes permanent damage to the kidneys in ~6% of patients with elevated blood pressure. As many as 27% of patients with end-stage kidney disease have hypertension as a primary cause, and it is the second most common cause of ESRD after diabetic nephropathy. *Hypertensive nephrosclerosis* is fivefold more frequent in African Americans than whites. Risk alleles associated with *APOL1,* a functional gene for apolipoprotein L1 expressed in podocytes, substantially explain the increased burden of ESRD among African Americans. Associated risk factors for progression to end-stage kidney disease include increased age, male gender, race, smoking, hypercholesterolemia, duration of hypertension, low birth weight, and preexisting renal injury. Kidney biopsies in patients with hypertension, microhematuria, and moderate proteinuria demonstrate arteriolosclerosis, chronic nephrosclerosis, and interstitial fibrosis in the absence of immune deposits **(see Fig. A4-21)**. Based on a careful history, physical examination, urinalysis, and some serologic testing, the diagnosis of chronic nephrosclerosis is usually inferred without a biopsy. Recent studies suggest, in the absence of diabetes, adults with hypertension and cardiovascular risk factors benefit from achieving a systolic blood pressure <120 mmHg, compared to <140 mmHg. In the presence of kidney disease, most patients begin antihypertensive therapy with two drugs, classically a thiazide diuretic and an ACE inhibitor; most will require three drugs. There is strong evidence in African Americans with hypertensive nephrosclerosis that therapy initiated with an ACE inhibitor can slow the rate of decline in renal function independent of effects on systemic blood pressure. Malignant acceleration of hypertension complicates the course of chronic nephrosclerosis, particularly in the setting of scleroderma or cocaine use **(see Fig. A4-24)**. The hemodynamic stress of malignant hypertension leads to fibrinoid necrosis of

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small blood vessels, thrombotic microangiography, a nephritic urinalysis, and acute renal failure. In the setting of renal failure, chest pain, or papilledema, the condition is treated as a hypertensive emergency.

CHOLESTEROL EMBOLI

Aging patients with clinical complications from atherosclerosis sometimes shower cholesterol crystals into the circulation, either spontaneously or, more commonly, following an endovascular procedure with manipulation of the aorta or with use of systemic anticoagulation. Spontaneous emboli may shower acutely or shower subacutely and somewhat more silently. Irregular emboli trapped in the microcirculation produce ischemic damage that induces an inflammatory reaction. Depending on the location of the atherosclerotic plaques releasing these cholesterol fragments, one may see cerebral transient ischemic attacks; livedo reticularis in the lower extremities; Hollenhorst plaques in the retina with visual field cuts; necrosis of the toes; and acute glomerular capillary injury leading to FSGS sometimes associated with hematuria, mild proteinuria, and loss of renal function, which typically progresses over a few years. Occasional patients have fever, eosinophilia, or eosinophiluria. A skin biopsy of an involved area may be diagnostic. Since tissue fixation dissolves the cholesterol, one typically sees only residual, biconvex clefts in involved vessels **(see Fig. A4-22)**. There is no therapy to reverse embolic occlusions, and steroids do not help. Controlling blood pressure and lipids and cessation of smoking are usually recommended for prevention.

SICKLE CELL DISEASE

Although individuals with SA-hemoglobin are usually asymptomatic, most will gradually develop hyposthenuria due to subclinical infarction of the renal medulla, thus predisposing them to volume depletion. There is an unexpectedly high prevalence of sickle trait among dialysis patients who are African American. Patients with homozygous SS-sickle cell disease and less commonly SC-sickle cell disease develop chronic vaso-occlusive disease in many organs. Polymers of deoxygenated SS-hemoglobin distort the shape of red blood cells. These cells attach to endothelia and obstruct small blood vessels, producing frequent and painful sickle cell crises over time. Early changes in the kidney include glomerular hyperfiltration, hyposthenuria, micro- or macroscopic hematuria, and microalbuminuria. Later changes can include papillary necrosis, renal infarction and proteinuria, and most commonly, FSGS on renal biopsy and rarely MPGN. Vessel occlusions in the kidney produce glomerular hypertension, FSGS, interstitial nephritis, and renal infarction associated with hyposthenuria, microscopic hematuria, and even gross hematuria; some patients also present with MPGN. Renal function can be overestimated due to the increased tubular secretion of creatinine seen in many patients with SS-sickle cell. By the second or third decade of life, persistent vaso-occlusive disease in the kidney leads to varying degrees of renal failure. Their prognosis on dialysis is poor, and anemia management with erythropoiesis-stimulating agents is complicated. Treatment is directed to reducing the frequency of painful crises and administering ACE inhibitors and hydroxyurea in the hope of delaying a progressive decline in renal function. In sickle cell patients undergoing renal transplantation, renal graft survival is comparable to African Americans in the general transplant population.

THROMBOTIC MICROANGIOPATHIES

Thrombotic thrombocytopenic purpura (TTP), Shiga toxin–mediated *hemolytic-uremic syndrome* (HUS), and complement-mediated HUS represent a spectrum of thrombotic microangiopathies (TMAs). TTP and HUS share the general features of idiopathic thrombocytopenic purpura, hemolytic anemia, fever, renal failure, and neurologic disturbances. Clinically, when patients, particularly children, have evidence of renal injury, HUS is suspected, and in adults with neurologic disease, TTP is suspected. On examination of kidney tissue, there is evidence of *glomerular capillary endotheliosis* associated with platelet thrombi, damage to the capillary wall, and formation of fibrin material in and around glomeruli **(see Fig. A4-23)**. These tissue findings are similar to what is seen in preeclampsia/HELLP (*h*emolysis, *e*levated *l*iver enzymes, and *l*ow *p*latelet count syndrome), malignant hypertension, and the antiphospholipid syndrome. TMA is also seen postpartum (and may be complement mediated); with the use of oral contraceptives or quinine; in renal transplant patients given OKT3 for rejection; in patients taking the calcineurin inhibitors cyclosporine and tacrolimus; in patients taking the antiplatelet agents ticlopidine and clopidogrel; and following HIV infection. The implicated drug should be discontinued.

Shiga toxin–mediated HUS is caused by a toxin released by *Escherichia coli* 0157:H7 and occasionally by *Shigella dysenteriae*. This Shiga toxin (verotoxin) directly injures endothelia, enterocytes, and renal cells, causing apoptosis, platelet clumping, and intravascular hemolysis by binding to the glycolipid receptors (Gb3). These receptors are more abundant along endothelia in children compared to adults. Shiga toxin also inhibits the endothelial production of ADAMTS13. In familial cases of adult TTP, there is a genetic deficiency of the ADAMTS13 metalloprotease that cleaves large multimers of von Willebrand's factor. Absent ADAMTS13, these large multimers cause platelet clumping and intravascular hemolysis. An antibody to ADAMTS13 is found in many sporadic cases of adult TTP. Patients can be tested for ADAMTS13 activity, and if low, the presence of antibodies to ADAMTS13 distinguishes the deficiency from the immune-mediated disease. Complement-mediated TMA/HUS is a rare hereditary deficiency of one of the regulatory proteins that restrict the activation of the alternate complement pathway and can present in children or adults often preceded by an infection. The treatment of adult TTP with ADAMTS13 antibodies is daily plasmapheresis, which can be lifesaving. Plasmapheresis with fresh frozen plasma is given until the platelet count rises, but in relapsing patients, it normally is continued well after the platelet count improves. There is an anecdotal role in relapsing patients for using splenectomy, steroids, immunosuppressive drugs, bortezomib, or rituximab. Patients without antibodies and a genetic deficiency of ADAMTS13 production can be treated with fresh frozen plasma alone. Patients with Shiga toxin–mediated HUS are not given antibiotics and are treated with supportive care because antibiotics are thought to accelerate the release of the toxin and the diarrhea is usually self-limited. Patients with complement-mediated TMA/HUS are treated with eculizumab, an anticomplement therapy.

ANTIPHOSPHOLIPID ANTIBODY SYNDROME
(SEE CHAP. 357)

GLOBAL CONSIDERATIONS

INFECTIOUS DISEASE–ASSOCIATED SYNDROMES

A number of infectious diseases will injure the glomerular capillaries as part of a systemic reaction producing an immune response or from direct infection of renal tissue. Evidence of this immune response is collected by glomeruli in the form of immune deposits that damage the kidney, producing moderate proteinuria and hematuria. A high prevalence of many of these infectious diseases in developing countries results in infection-associated renal disease being the most common cause of glomerulonephritis in many parts of the world.

Poststreptococcal Glomerulonephritis This form of glomerulonephritis is one of the classic complications of streptococcal infection. The discussion of this disease can be found earlier, in the section "Acute Nephritic Syndromes."

Subacute Bacterial Endocarditis Renal injury from persistent bacteremia absent the continued presence of a foreign body, regardless of cause, is treated presumptively as if the patient has endocarditis. The discussion of this disease can be found earlier, in the section "Acute Nephritic Syndromes."

Human Immunodeficiency Virus Renal disease is an important complication of HIV disease. The risk of development of ESRD is much higher in HIV-infected African Americans than in HIV-infected whites. About 50% of HIV-infected patients with kidney disease have HIV-associated nephropathy (HIVAN) on biopsy. The lesion

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in HIVAN is FSGS, characteristically revealing a collapsing glomerulopathy **(see Fig. A4-3)** with visceral epithelial cell swelling, microcystic dilatation of renal tubules, and tubuloreticular inclusion. Renal epithelial cells express replicating HIV virus, but host immune responses also play a role in the pathogenesis. HIVAN develops almost exclusively in patients of black race origin, linked to *APOL1* polymorphisms. HIV immune complex kidney disease (HIVICK) is a group of immune complex–mediated glomerular lesions seen in HIV patients that, on biopsy, can look like a constellation of other glomerular lesions, including postinfectious glomerulonephritis, MGN, MPGN, DPGN, MCD, and IgA nephropathy. The HIVICK effect is a complication of active HIV viremia.

HIV patients with FSGS typically present with nephrotic-range proteinuria and hypoalbuminemia, but unlike patients with other etiologies for nephrotic syndrome, they do not commonly have hypertension, edema, or hyperlipidemia. Renal ultrasound also reveals large, echogenic kidneys despite the finding that renal function in some patients declines rapidly. Treatment with inhibitors of the renin-angiotensin system decreases the proteinuria. Effective antiretroviral therapy benefits both the patient and the kidney and improves survival of HIV-infected patients with HIVAN and, in some cases, HIVICK-associated chronic kidney disease or ESRD. In HIV-infected patients not yet on therapy, the presence of HIVAN is an indication to initiate therapy. Following the introduction of antiretroviral therapy, survival on dialysis for the HIV-infected patient has improved dramatically. Renal transplantations in HIV-infected patients without detectable viral loads or histories of opportunistic infections provide a better survival benefit over dialysis. Following transplantation, patient and graft survival are similar to the general transplant population despite frequent rejections.

Hepatitis B and C Typically, infected patients present with microscopic hematuria, nonnephrotic or nephrotic-range proteinuria, and hypertension. There is a close association between hepatitis B infection and polyarteritis nodosa, with vasculitis appearing generally in the first 6 months following infection. Renal manifestations include renal artery aneurysms, renal infarction, and ischemic scars. Alternatively, the hepatitis B carrier state can produce an MGN with predominant IgG1 deposition that is more common in children than adults or MPGN that is more common in adults than in children. Renal histology is indistinguishable from idiopathic MGN or MPGN. Viral antigens, most commonly HBeAG, are found in the renal deposits. Cryoglobulinemic glomerulonephritis has also been reported. Treatment is with antiviral agents. Children have a better prognosis than adults.

Up to 30% of patients with chronic hepatitis C infection have some renal manifestations. Patients often present with type II mixed cryoglobulinemia, nephrotic syndrome, microscopic hematuria, abnormal liver function tests, depressed C_3 levels, anti–hepatitis C virus (HCV) antibodies, and viral RNA in the blood. The renal lesions most commonly seen, in order of decreasing frequency, are cryoglobulinemic glomerulonephritis, MGN, and MPGN, but polyarteritis nodosa (PAN), IgA nephropathy, and FSGS have been reported. With the availability of direct-acting antivirals, which can achieve a viral remission in >95% of patients, the prevalence of glomerular disease in HCV patients should decline. These drugs are currently the treatment of choice for patients with HCV-related MPGN or PAN.

Other Viruses Other viral infections are occasionally associated with glomerular lesions, but cause and effect are not well established. These viral infections and their respective glomerular lesions include cytomegalovirus producing MPGN or FSGS; influenza and anti-GBM disease; measles-associated endocapillary proliferative glomerulonephritis, with measles antigen in the capillary loops and mesangium; parvovirus causing mild proliferative or mesangioproliferative glomerulonephritis or FSGS; mumps and mesangioproliferative glomerulonephritis; Epstein-Barr virus producing MPGN, diffuse proliferative nephritis, or IgA nephropathy; dengue hemorrhagic fever causing endocapillary proliferative glomerulonephritis; Hanta virus and mesangial proliferative glomerulonephritis; and coxsackievirus producing *focal glomerulonephritis* or DPGN.

Syphilis Secondary syphilis, with rash and constitutional symptoms, develops weeks to months after the chancre first appears and occasionally presents with the nephrotic syndrome from MGN caused by subepithelial immune deposits containing treponemal antigens. Other lesions have also rarely been described, including interstitial syphilitic nephritis. The diagnosis is confirmed with nontreponemal and treponemal tests for *Treponema pallidum*. The renal lesion responds to treatment with penicillin or an alternative drug, if allergic. Additional testing for other sexually transmitted diseases is an important part of disease management.

Leprosy Despite aggressive eradication programs, new cases of leprosy appear primarily in developing countries. The diagnosis is best made in patients with multiple skin lesions accompanied by sensory loss in affected areas, using skin smears showing paucibacillary or multibacillary infection (WHO criteria). Leprosy is caused by infection with *Mycobacterium leprae* and can be classified by Ridley-Jopling criteria into various types: tuberculoid, borderline tuberculoid, mid-borderline and borderline lepromatous, and lepromatous. Renal involvement in leprosy is related to the quantity of bacilli in the body, and the kidney is one of the target organs during splanchnic localization. In some series, all cases with borderline lepromatous and lepromatous types of leprosy have various forms of renal involvement including FSGS, mesangioproliferative glomerulonephritis, or renal amyloidosis; much less common are the renal lesions of DPGN and MPGN. Treatment of the infection with multidrug therapy can reduce the incidence of renal disease or produce remission of the renal disease.

Malaria There are 300–500 million incident cases of malaria each year worldwide, and the kidney is commonly involved. Glomerulonephritis is due to immune complexes containing malarial antigens that are implanted in the glomerulus. In malaria from *P. falciparum*, mild proteinuria is associated with subendothelial deposits, mesangial deposits, and mesangioproliferative glomerulonephritis that usually resolve with treatment. In quartan malaria from infection with *Plasmodium malariae*, children are more commonly affected and renal involvement is more severe. Transient proteinuria and microscopic hematuria can resolve with treatment of the infection. However, resistant nephrotic syndrome with progression to renal failure over 3–5 years does happen, as <50% of patients respond to steroid therapy. Affected patients with nephrotic syndrome have thickening of the glomerular capillary walls, with subendothelial deposits of IgG, IgM, and C_3 associated with a sparse membranoproliferative lesion. The rare mesangioproliferative glomerulonephritis reported with *Plasmodium vivax* or *Plasmodium ovale* typically has a benign course. Acute kidney injury can often complicate these glomerulopathies.

Schistosomiasis Schistosomiasis affects >300 million people worldwide and primarily involves the urinary and gastrointestinal tracts. Glomerular involvement varies with the specific strain of schistosomiasis; *Schistosoma mansoni* is most commonly associated with clinical renal disease, and the glomerular lesions can be classified as follows: class I is a *mesangioproliferative glomerulonephritis*; class II is an *extracapillary proliferative glomerulonephritis*; class III is a *membranoproliferative glomerulonephritis*; class IV is a *focal segmental glomerulonephritis*; and class V is *amyloidosis*. Classes I–II often remit with treatment of the infection, but class III and IV lesions are associated with IgA immune deposits and progress despite antiparasitic and/or immunosuppressive therapy.

Other Parasites Renal involvement with toxoplasmosis infections is rare. When it occurs, patients present with nephrotic syndrome and have a histologic picture of MPGN. Fifty percent of patients with leishmaniasis will have mild to moderate proteinuria and microscopic hematuria, but renal insufficiency is rare. Acute DPGN, MGN, and mesangioproliferative glomerulonephritis have all been observed on biopsy. Filariasis and trichinosis are caused by nematodes and are sometimes associated with glomerular injury presenting with proteinuria, hematuria, and a variety of histologic lesions that typically resolve with eradication of the infection.

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315 Polycystic Kidney Disease and Other Inherited Disorders of Tubule Growth and Development

Jing Zhou, Martin R. Pollak

The polycystic kidney diseases are a group of genetically heterogeneous disorders and a leading cause of kidney failure. The autosomal dominant form of polycystic kidney disease (ADPKD) is the most common life-threatening monogenic disease, affecting 12 million people worldwide. The autosomal recessive form of polycystic kidney disease (ARPKD) is rarer but affects the pediatric population. Kidney cysts are often seen in a wide range of syndromic diseases. Recent studies have shown that defects in the structure or function of the primary cilia may underlie this group of genetic diseases collectively termed *ciliopathies* (Table 315-1).

AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE

Etiology and Pathogenesis (Fig. 315-1) ADPKD is characterized by progressive formation of epithelial lined cysts in the kidney. Although cysts occur in only 5% of the tubules in the kidney, the enormous growth of these cysts ultimately leads to the loss of normal surrounding tissues and loss of renal function. The cellular defects in ADPKD that have been known for a long time are increased cell proliferation and fluid secretion, decreased cell differentiation, and abnormal extracellular matrix. ADPKD is caused by mutations in *PKD1* and *PKD2*, which, respectively, code for polycystin-1 (PC1) and polycystin-2 (PC2). PC1 is a large 11-transmembrane protein that functions like a G-protein coupled receptor. PC2 is a calcium-permeable six-transmembrane protein that structurally belongs to the transient receptor potential (TRP) cation channel family. PC1 and PC2 are widely expressed in almost all tissues and organs. PC1 expression is high in development and low in the adult, whereas PC2 expression is relatively constant. PC1/2 are found on the primary cilium, a hairlike structure present on the apical membrane of a cell, in addition to the cell membranes and cell–cell junctions of tubular epithelial cells. Defects in the primary cilia are linked to a wide spectrum of human diseases, collectively termed *ciliopathies*. The most common phenotype shared by many ciliopathies is kidney cysts. PC1 and PC2 bind to each other via their respective C-terminal tails to form a receptor-channel complex and regulate each other's function. Recent evidence suggests a 1:3 stoichiometry for PC1:PC2 in the PC1/2 channel complex. The PC1/2 protein complex serves as a mechanosensor or chemical sensor and regulates calcium and G-protein signaling. The PC1/2 protein complex may also directly regulate a number of cellular functions, including the cell cycle, the actin cytoskeleton, planar cell polarity (PCP), and cell migration. This protein complex has also been implicated in regulating a number of signaling pathways, including Wnt, mammalian target of rapamycin (mTOR), STAT3, cMET, phosphoinositide 3-kinase (PI3K/Akt), G protein–coupled receptor (GPCR), and epidermal growth factor receptor (EGFR), as well as in the localization and activity of cystic fibrosis transmembrane conductance regulator (CFTR). One hypothesis is that loss of ciliary function of PC1 and PC2 leads to aberrant calcium signaling and a subsequent increase of adenylyl cyclase activity and decrease of phosphodiesterase activity, which, in turn, causes increased cellular cAMP. Increased cAMP promotes protein kinase A activity, among other effectors, and, in turn, leads to cyst growth by promoting proliferation and fluid secretion of cyst-lining cells through chloride and aquaporin channels in ADPKD kidneys.

ADPKD is inherited as an autosomal dominant trait with complete penetrance, but variable expressivity. The disease affects all ethnic groups worldwide with an estimated prevalence of 1:1000 to 1:400. Only half of the patients with ADPKD are clinically diagnosed during their lifetimes. ADPKD is genetically heterogeneous. The first disease gene (*PKD1*) was localized to the region of the alpha-globin gene on chromosome 16p13 in 1985, and a second disease gene (*PKD2*) locus was mapped to chromosome 4q21-q23 in 1993. Mutations of *PKD1* and *PKD2* are responsible for ~85% and ~15% of ADPKD cases, respectively. However, patients with *PKD2* mutations may be >15% because they tend to have milder clinical disease and, as a result, are underdiagnosed. Embryonic lethality of *Pkd1* and *Pkd2* knockout mice suggest human homozygotes may be lethal, thus not clinically recognized.

PKD1 is comprised of 46 exons occupying ~52 kb of genomic DNA. It produces a ~14 kb transcript that encodes polycystin-1, a protein of ~4300 amino acids. A feature of the *PKD1* gene is that the 5' three-quarters of *PKD1* have been duplicated at six other sites on chromosome 16p, and many of them produce mRNA transcripts, which provides a major challenge for genetic analysis of the duplicated region. *PKD2* is a single-copy gene with 15 exons producing a ~5.3 kb mRNA transcript that encodes polycystin-2, a protein of 968 amino acids. Two additional genes, *GANAB* and *DNAJB11,* have been found in patients with autosomal dominant form of polycystic kidney disease. The *GANAB* gene encodes the glucosidase IIa subunit and the *DNAJB11* gene produces a cofactor of BiP, a key chaperone in the endoplasmic reticulum controlling folding, trafficking, and degradation of secreted and membrane proteins. Both proteins appear to affect PC1 trafficking. However, these mutations have only been found in a very small number of families.

In ADPKD patients, every cell carries a germline mutant allele of either *PKD1* or *PKD2*. However, cysts develop in only a small fraction of the nephrons. Cysts are thought to originate from clonal growth of single cells that have received a somatic "second hit" mutation in the "normal" allele of the *PKD1* or *PKD2* gene. Accumulating evidence in mouse models now shows that partial loss of function of the second allele of *Pkd1* in a proliferative environment is sufficient for cystogenesis, suggesting that a critical amount of *PKD1* is needed in a cell. Somatic inactivation of the second allele of *Pkd1* in adult mice results in very slow onset of cyst development in the kidney, but a "third hit" such as an additional genetic or epigenetic event, the inactivation of a growth suppressor gene, the activation of a growth promoting gene(s), or an event such as renal injury that activates the developmental program, may promote rapid cyst formation.

Clinical Manifestations ADPKD is characterized by the progressive bilateral formation of renal cysts. Focal renal cysts are typically detected in affected subjects aged <30 years. Hundreds to thousands of cysts are usually present in the kidneys of most patients in the fifth decade (Fig. 315-2). Enlarged kidneys can each reach a four-fold increase in length and weigh up to 20 times the normal weight.